

EDEXCEL INTERNATIONAL A LEVEL

# WME01 Mechanics 1 Expertise Questions

Questions 6 and above, grouped by topic with cross-topic placements where useful.

**107**

topic placements

**WME01**

Mechanics 1

**Question  
bank**standalone IAL  
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

**M1**

**TOPIC**

# **Quantities, Units & Modelling**

12 marks

WME01/01 OCTOBER 2019

## Question 6

Quantities, Units & Modelling

Also in Quantities, Units & Modelling

Primary: Kinematics Graphs

6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time  $t = 0$  seconds and moves with uniform acceleration  $0.8 \text{ m s}^{-2}$  for  $T$  seconds, reaching a speed of  $V \text{ m s}^{-1}$ . She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (a) Sketch a speed-time graph for the model of the motion of the athlete from the instant when she starts to the instant when she crosses the finishing line. (2)
- (b) Write down an expression for  $V$  in terms of  $T$ . (1)
- (c) Show that  $T^2 - kT + 500 = 0$ , where  $k$  is a constant to be found. (4)
- (d) Hence find the value of  $T$ , justifying your answer carefully. (3)
- (e) Considering your speed-time graph or otherwise, state two ways, apart from including air resistance, in which the model could be made to be more realistic. (2)

10 marks

WME01/01 MAY/JUNE 2021

Quantities, Units & Modelling

## Question 7

Also in Quantities, Units & Modelling

Primary: Moments

7.

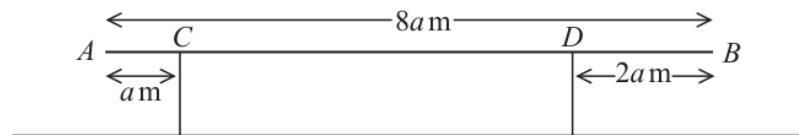


Figure 2

A non-uniform beam  $AB$ , of mass  $60$  kg and length  $8a$  metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at  $C$ , where  $AC = a$  metres and the other support is at  $D$ , where  $DB = 2a$  metres, as shown in Figure 2.

The magnitude of the normal reaction between the beam and the support at  $D$  is three times the magnitude of the normal reaction between the beam and the support at  $C$ .

By modelling the beam as a non-uniform rod whose centre of mass is at a distance  $x$  metres from  $A$ ,

- (a) find an expression for  $x$  in terms of  $a$ .

(5)

A box of mass  $M$  kg is placed on the beam at  $E$ , where  $AE = 2a$  metres.

The beam remains in equilibrium in a horizontal position.

The magnitude of the normal reaction between the beam and the support at  $C$  is now equal to the magnitude of the normal reaction between the beam and the support at  $D$ .

By modelling the box as a particle,

- (b) find the value of  $M$ .

(5)



8 marks

WME01/01 JANUARY 2023

Quantities, Units & Modelling

## Question 6

Also in Quantities, Units & Modelling

Primary: Forces

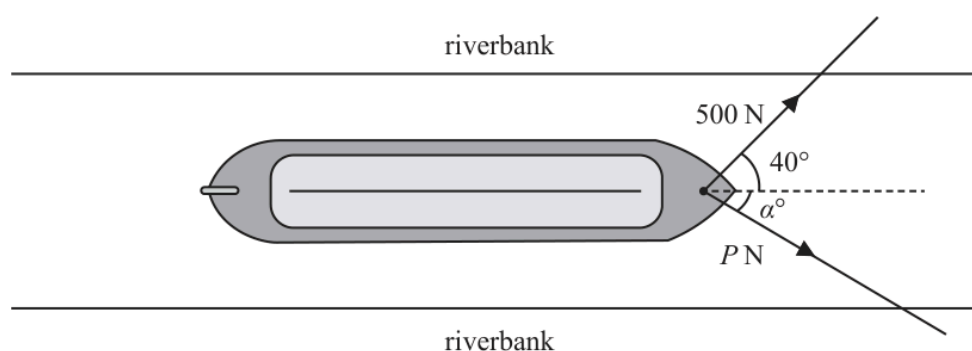


Figure 3

A boat is pulled along a river at a constant speed by two ropes.

The banks of the river are parallel and the boat travels horizontally in a straight line, parallel to the riverbanks.

- The tension in the first rope is 500 N acting at an angle of  $40^\circ$  to the direction of motion, as shown in Figure 3.
- The tension in the second rope is  $P$  newtons, acting at an angle of  $\alpha^\circ$  to the direction of motion, also shown in Figure 3.
- The resistance to motion of the boat as it moves through the water is a constant force of magnitude 900 N

The boat is modelled as a particle. The ropes are modelled as being light and lying in a horizontal plane.

Use the model to find

- (i) the value of  $\alpha$
- (ii) the value of  $P$

(8)

7 marks

WME01/01 JANUARY 2023

## Question 7

Quantities, Units & Modelling

Also in Quantities, Units & Modelling

Primary: Newton's Second Law

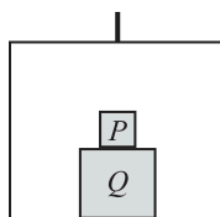


Figure 4

A simple lift operates by means of a vertical cable which is attached to the top of the lift.

The lift has mass  $m$

A box  $Q$  is placed on the floor of the lift.

A box  $P$  is placed directly on top of box  $Q$ , as shown in Figure 4.

The cable is modelled as being light and inextensible and air resistance is modelled as being negligible.

The tension in the cable is  $\frac{42mg}{5}$

The lift and its contents move vertically upwards with acceleration  $\frac{2g}{5}$

Using the model,

(a) find, in terms of  $m$ , the combined mass of boxes  $P$  and  $Q$

(4)

During the motion of the lift, the force exerted on box  $P$  by box  $Q$  is  $\frac{14mg}{5}$

Using the model,

(b) find, in terms of  $m$ , the mass of box  $P$

(3)

**TOPIC**

# Working with Vectors

## Question 7

## Working with Vectors

7. Two forces,  $\mathbf{F}$  and  $\mathbf{G}$ , act on a particle. The force  $\mathbf{F}$  has magnitude 4N and acts in a direction with a bearing of  $120^\circ$  and the force  $\mathbf{G}$  has magnitude 6N and acts due north.

Given that  $\mathbf{P} = 2\mathbf{F} + \mathbf{G}$ , find

- (i) the magnitude of  $\mathbf{P}$
- (ii) the direction of  $\mathbf{P}$ , giving your answer as a bearing to the nearest degree.

(7)

## Question 8

## Working with Vectors

8. [In this question, the horizontal unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

Two speedboats,  $A$  and  $B$ , are each moving with constant velocity. The velocity of  $A$  is  $20 \text{ km h}^{-1}$  due west and the velocity of  $B$  is  $40 \text{ km h}^{-1}$  on a bearing of  $150^\circ$ . The boats are modelled as particles.

At noon, the position vector of  $A$  is  $60\mathbf{i} \text{ km}$  and  $B$  is at the origin  $O$ . At time  $t$  hours after noon, the position vector of  $A$  is  $\mathbf{r} \text{ km}$  and the position vector of  $B$  is  $\mathbf{s} \text{ km}$ .

- (a) Find the velocity of  $B$  in the form  $(p\mathbf{i} + q\mathbf{j}) \text{ km h}^{-1}$  (3)
- (b) Find expressions for  $\mathbf{r}$  and  $\mathbf{s}$  in terms of  $t$ ,  $\mathbf{i}$  and  $\mathbf{j}$ . (3)
- (c) Find the time, to the nearest minute, at which the distance between the boats is the same as it was at noon. (8)

## Question 6

## Working with Vectors

6. A force  $\mathbf{F}$  is given by  $\mathbf{F} = (10\mathbf{i} + \mathbf{j})\text{N}$ .

(a) Find the exact value of the magnitude of  $\mathbf{F}$ .

(2)

(b) Find, in degrees, the size of the angle between the direction of  $\mathbf{F}$  and the direction of the vector  $(\mathbf{i} + \mathbf{j})$ .

(4)

The resultant of the force  $\mathbf{F}$  and the force  $(-15\mathbf{i} + a\mathbf{j})\text{N}$ , where  $a$  is a constant, is parallel to, but in the opposite direction to, the vector  $(2\mathbf{i} - 3\mathbf{j})$ .

(c) Find the value of  $a$ .

(5)

## Question 6

## Working with Vectors

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

At time  $t = 0$ , Agatha is at the point with position vector  $(11\mathbf{i} + 11\mathbf{j})\text{m}$  and Brionie is at the point with position vector  $(7\mathbf{i} + 16\mathbf{j})\text{m}$ . The position vectors are given relative to the door,  $O$ , and  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal perpendicular unit vectors.

Agatha skates with constant velocity  $(3\mathbf{i} - \mathbf{j})\text{ms}^{-1}$

Brionie skates with constant velocity  $(4\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$

- (a) Find the position vector of Agatha at time  $t$  seconds. (2)

At time  $t = 6$  seconds, Agatha passes through the point  $P$ .

- (b) Show that Brionie also passes through  $P$  and find the value of  $t$  when this occurs. (4)

At time  $t$  seconds, Agatha is at the point  $A$  and Brionie is at the point  $B$ .

- (c) Show that  $\overrightarrow{AB} = [(t - 4)\mathbf{i} + (5 - t)\mathbf{j}]\text{m}$  (2)

- (d) Find the distance between the two girls when they are closest together. (4)

## Question 8

## Working with Vectors

8. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

At 7 am a ship leaves a port and moves with constant velocity. The position vector of the port is  $(-2\mathbf{i} + 9\mathbf{j})$  km.

At 7.36 am the ship is at the point with position vector  $(4\mathbf{i} + 6\mathbf{j})$  km.

- (a) Show that the velocity of the ship is  $(10\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$  (2)

- (b) Find the position vector of the ship  $t$  hours after leaving port. (2)

At 8.48 am a passenger on the ship notices that a lighthouse is due east of the ship.

At 9 am the same passenger notices that the lighthouse is now north east of the ship.

- (c) Find the position vector of the lighthouse. (4)

- (d) Find the position vector of the ship when it is due south of the lighthouse. (4)

(Total 12 marks)



## Question 6

## Working with Vectors

6. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors.]

A particle  $P$  of mass  $2\text{ kg}$  moves under the action of two forces,  $(p\mathbf{i} + q\mathbf{j})\text{ N}$  and  $(2q\mathbf{i} + p\mathbf{j})\text{ N}$ , where  $p$  and  $q$  are constants.

Given that the acceleration of  $P$  is  $(\mathbf{i} - \mathbf{j})\text{ m s}^{-2}$

(a) find the value of  $p$  and the value of  $q$ .

(5)

(b) Find the size of the angle between the direction of the acceleration and the vector  $\mathbf{j}$ .

(2)

At time  $t = 0$ , the velocity of  $P$  is  $(3\mathbf{i} - 4\mathbf{j})\text{ m s}^{-1}$

At  $t = T$  seconds,  $P$  is moving in the direction of the vector  $(11\mathbf{i} - 13\mathbf{j})$ .

(c) Find the value of  $T$ .

(5)

## Question 8

## Working with Vectors

8. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

A ship  $A$  moves with constant velocity  $(3\mathbf{i} - 10\mathbf{j})\text{ km h}^{-1}$

At time  $t$  hours, the position vector of  $A$  is  $\mathbf{r}$  km.

At time  $t = 0$ ,  $A$  is at the point with position vector  $(13\mathbf{i} + 5\mathbf{j})\text{ km}$ .

- (a) Find  $\mathbf{r}$  in terms of  $t$ .

(2)

Another ship  $B$  moves with constant velocity  $(15\mathbf{i} + 14\mathbf{j})\text{ km h}^{-1}$

At time  $t = 0$ ,  $B$  is at the point with position vector  $(3\mathbf{i} - 5\mathbf{j})\text{ km}$ .

- (b) Show that, at time  $t$  hours,

$$\overrightarrow{AB} = [(12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}] \text{ km}$$

(4)

Given that the two ships do not change course,

- (c) find the shortest distance between the two ships,

(6)

- (d) find the bearing of ship  $B$  from ship  $A$  when the ships are closest.

(2)

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(Total 14 marks)

## Question 6

### Working with Vectors

6. A particle  $P$  is moving with constant acceleration.

At time  $t = 1$  second,  $P$  has velocity  $(-\mathbf{i} + 4\mathbf{j})\text{ms}^{-1}$

At time  $t = 4$  seconds,  $P$  has velocity  $(5\mathbf{i} - 8\mathbf{j})\text{ms}^{-1}$

Find the speed of  $P$  at time  $t = 3.5$  seconds.

(6)

## Question 8

## Working with Vectors

8. [In this question,  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

Two boats,  $P$  and  $Q$ , are moving with constant velocities.

The velocity of  $P$  is  $15\mathbf{i} \text{ m s}^{-1}$  and the velocity of  $Q$  is  $(20\mathbf{i} - 20\mathbf{j}) \text{ m s}^{-1}$

- (a) Find the direction in which  $Q$  is travelling, giving your answer as a bearing.

(2)

The boats are modelled as particles.

At time  $t = 0$ ,  $P$  is at the origin  $O$  and  $Q$  is at the point with position vector  $200\mathbf{j} \text{ m}$ .

At time  $t$  seconds, the position vector of  $P$  is  $\mathbf{p} \text{ m}$  and the position vector of  $Q$  is  $\mathbf{q} \text{ m}$ .

- (b) Show that

$$\overrightarrow{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m} \quad (5)$$

- (c) Find the bearing of  $P$  from  $Q$  when  $t = 10$

(2)

- (d) Find the distance between  $P$  and  $Q$  when  $Q$  is north east of  $P$

(5)

- (e) Find the times when  $P$  and  $Q$  are 200 m apart.

(3)

(Total 17 marks)

## Question 6

## Working with Vectors

6. [In this question,  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors.]

A particle  $A$  of mass  $0.5\text{ kg}$  is at rest on a smooth horizontal plane.

At time  $t = 0$ , two forces,  $\mathbf{F}_1 = (-3\mathbf{i} + 2\mathbf{j})\text{ N}$  and  $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})\text{ N}$ , where  $p$  and  $q$  are constants, are applied to  $A$ .

Given that  $A$  moves in the direction of the vector  $(\mathbf{i} - 2\mathbf{j})$ ,

(a) show that  $2p + q - 4 = 0$  (4)

Given that  $p = 5$

(b) find the speed of  $A$  at time  $t = 4$  seconds. (5)

## Question 8

## Working with Vectors

8. [In this question,  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

Two ships,  $A$  and  $B$ , are moving with constant velocities.

The velocity of  $A$  is  $(3\mathbf{i} + 12\mathbf{j})\text{ km h}^{-1}$  and the velocity of  $B$  is  $(p\mathbf{i} + q\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of  $A$ .

(2)

The ships are modelled as particles.

At 12 noon,  $A$  is at the point with position vector  $(-9\mathbf{i} + 6\mathbf{j})\text{ km}$  and  $B$  is at the point with position vector  $(16\mathbf{i} + 6\mathbf{j})\text{ km}$ .

At time  $t$  hours after 12 noon,

$$\overrightarrow{AB} = [(25 - 12t)\mathbf{i} - 9t\mathbf{j}]\text{ km}$$

- (b) Find the value of  $p$  and the value of  $q$ .

(7)

- (c) Find the bearing of  $A$  from  $B$  when the ships are 15 km apart, giving your answer to the nearest degree.

(7)

(Total 16 marks)

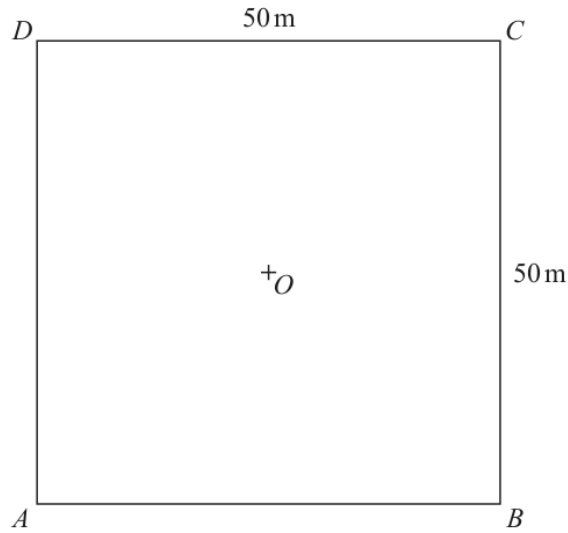
WME01/01 MAY/JUNE 2023

9 marks

## Question 8

Working with Vectors

8.



**Figure 5**

A square floor space  $ABCD$ , with centre  $O$ , is modelled as a flat horizontal surface measuring 50 m by 50 m, as shown in Figure 5.

The horizontal unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are in the direction of  $\overrightarrow{AB}$  and  $\overrightarrow{AD}$  respectively.

All position vectors are given relative to  $O$ .

A small robot  $R$  is programmed to travel across the floor at a constant velocity.

- At time  $t = 0$ ,  $R$  is at the point with position vector  $(-2\mathbf{i} + \mathbf{j})\text{ m}$
- At time  $t = 11\text{ s}$ ,  $R$  is at the point with position vector  $(9\mathbf{i} + 23\mathbf{j})\text{ m}$
- At time  $t$  seconds, the position vector of  $R$  is  $\mathbf{r}$  metres

(a) Find, in terms of  $t$ ,  $\mathbf{i}$  and  $\mathbf{j}$ , an expression for  $\mathbf{r}$

**(3)**

A second robot  $S$  is at the point  $C$ .

- At time  $t = 0$ ,  $S$  leaves  $C$  and moves with constant velocity  $(-\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$
- At time  $t$  seconds, the position vector of  $S$  is  $\mathbf{s}$  metres

(b) Write down, in terms of  $t$ ,  $\mathbf{i}$  and  $\mathbf{j}$ , an expression for  $\mathbf{s}$

**(1)**

(c) Show that

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}] \text{ m}$$

**(2)**

(d) Find the time when the distance between  $R$  and  $S$  is a minimum.

**(3)**



## Question 6

## Working with Vectors

6. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

At 12:00, a ship  $P$  sets sail from a harbour with position vector  $(15\mathbf{i} + 36\mathbf{j})$  km.

At 12:30,  $P$  is at the point with position vector  $(20\mathbf{i} + 34\mathbf{j})$  km.

Given that  $P$  moves with constant velocity,

- (a) show that the velocity of  $P$  is  $(10\mathbf{i} - 4\mathbf{j}) \text{ km h}^{-1}$  (2)

At time  $t$  hours after 12:00, the position vector of  $P$  is  $\mathbf{p}$  km.

- (b) Find an expression for  $\mathbf{p}$  in terms of  $\mathbf{i}$ ,  $\mathbf{j}$  and  $t$ . (2)

A second ship  $Q$  is also travelling at a constant velocity.

At time  $t$  hours after 12:00, the position vector of  $Q$  is given by  $\mathbf{q}$  km, where

$$\mathbf{q} = (42 - 8t)\mathbf{i} + (9 + 14t)\mathbf{j}$$

Ships  $P$  and  $Q$  are modelled as particles.

If both ships maintained their course,

- (c) (i) verify that they would collide at 13:30  
(ii) find the position vector of the point at which the collision would occur. (4)

At 12:30  $Q$  changes speed and direction to avoid the collision.

Ship  $Q$  now travels due north with a constant speed of  $15 \text{ km h}^{-1}$

Ship  $P$  maintains the same constant velocity throughout.

- (d) Find the exact distance between  $P$  and  $Q$  at 14:30 (7)

## Question 7

## Working with Vectors

7. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

At midnight, a ship  $S$  is at the point with position vector  $(19\mathbf{i} + 22\mathbf{j})\text{ km}$

The ship travels with constant velocity  $(12\mathbf{i} - 16\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of  $S$ .

(2)

At time  $t$  hours after midnight, the position vector of  $S$  is  $\mathbf{s}$  km.

- (b) Find an expression for  $\mathbf{s}$  in terms of  $\mathbf{i}$ ,  $\mathbf{j}$  and  $t$ .

(2)

A lighthouse stands on a small rocky island. The lighthouse is modelled as being at the point with position vector  $(26\mathbf{i} + 15\mathbf{j})\text{ km}$ .

It is not safe for ships to be within 1.3 km of the lighthouse.

- (c) (i) Find the value of  $t$  when  $S$  is closest to the lighthouse.

- (ii) Hence determine whether it is safe for  $S$  to continue its course.

(7)

## Question 7

## Working with Vectors

7. [In this question, the horizontal unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

Two speedboats,  $A$  and  $B$ , are each moving with constant velocity.

- the velocity of  $A$  is  $40 \text{ km h}^{-1}$  due east
- the velocity of  $B$  is  $20 \text{ km h}^{-1}$  on a bearing of angle  $\alpha$  ( $0^\circ < \alpha < 90^\circ$ ), where  $\tan \alpha = \frac{4}{3}$

The boats are modelled as particles.

- (a) Find, in terms of  $\mathbf{i}$  and  $\mathbf{j}$ , the velocity of  $B$  in  $\text{km h}^{-1}$  (2)

At noon

- the position vector of  $A$  is  $20\mathbf{j}$  km
- the position vector of  $B$  is  $(10\mathbf{i} + 5\mathbf{j})$  km

At time  $t$  hours after noon

- the position vector of  $A$  is  $\mathbf{r}$  km, where  $\mathbf{r} = 20\mathbf{j} + 40t\mathbf{i}$
- the position vector of  $B$  is  $\mathbf{s}$  km

- (b) Find an expression for  $\mathbf{s}$  in terms of  $t$ ,  $\mathbf{i}$  and  $\mathbf{j}$ . (2)

- (c) Show that at time  $t$  hours after noon,

$$\overrightarrow{AB} = [(10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}] \text{ km} \quad (2)$$

- (d) Show that the boats will never collide. (3)

- (e) Find the distance between the boats when the bearing of  $B$  from  $A$  is  $225^\circ$  (4)

## Question 7

## Working with Vectors

7. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors.]

Two boats,  $A$  and  $B$ , are each moving with constant acceleration.

At time  $t$  hours after noon, boat  $A$  has velocity  $\mathbf{v}_A \text{ km h}^{-1}$ , where

$$\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$$

- (a) Find the magnitude of the acceleration of  $A$ . (2)

When  $t = 2$ , the velocity of  $B$  is  $(4\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

When  $t = 5$ , the velocity of  $B$  is  $(\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$

- (b) Find the acceleration of  $B$ , giving your answer in terms of  $\mathbf{i}$  and  $\mathbf{j}$ . (2)

- (c) Find the velocity of  $B$  at time  $t = 0$ , giving your answer in terms of  $\mathbf{i}$  and  $\mathbf{j}$ . (2)

At time  $T_1$  hours after noon, both boats are moving with the same speed.

- (d) Find the exact value of  $T_1$ . (4)

At time  $T_2$  hours after noon, both boats are moving in the same direction.

- (e) Show that  $3T_2^2 + pT_2 + q = 0$ , where  $p$  and  $q$  are integers to be found. (3)

## Question 6

## Working with Vectors

6. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

At 9 am, a ship  $S$  leaves a harbour and moves with constant velocity  $(10\mathbf{i} + 15\mathbf{j})\text{km h}^{-1}$

- (a) Find the exact speed of  $S$  (2)

The harbour is at the origin  $O$ .

At 9 am, a boat  $B$  leaves the point with position vector  $(50\mathbf{i} + 10\mathbf{j})\text{km}$  and moves with constant velocity  $(-8\mathbf{i} + 10\mathbf{j})\text{km h}^{-1}$

- (b) Find the direction in which  $B$  is moving, giving your answer as a bearing to the nearest degree. (3)

- (c) Show that at time  $t$  hours after 9 am,

$$\overrightarrow{SB} = [(50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}]\text{km} \quad (4)$$

- (d) Show that  $B$  and  $S$  do not collide. (3)

When  $B$  is north-east of  $S$ , the distance between  $S$  and  $B$  is  $d\text{km}$ .

- (e) Find the value of  $d$ . (4)

12 marks

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## Question 6

Working with Vectors

Also in Working with Vectors

Primary: Kinematics Graphs

6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time  $t = 0$  seconds and moves with uniform acceleration  $0.8 \text{ m s}^{-2}$  for  $T$  seconds, reaching a speed of  $V \text{ m s}^{-1}$ . She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (a) Sketch a speed-time graph for the model of the motion of the athlete from the instant when she starts to the instant when she crosses the finishing line. (2)
- (b) Write down an expression for  $V$  in terms of  $T$ . (1)
- (c) Show that  $T^2 - kT + 500 = 0$ , where  $k$  is a constant to be found. (4)
- (d) Hence find the value of  $T$ , justifying your answer carefully. (3)
- (e) Considering your speed-time graph or otherwise, state two ways, apart from including air resistance, in which the model could be made to be more realistic. (2)

8 marks

WME01/01 JANUARY 2023

## Question 6

Working with Vectors

Also in Working with Vectors

Primary: Forces

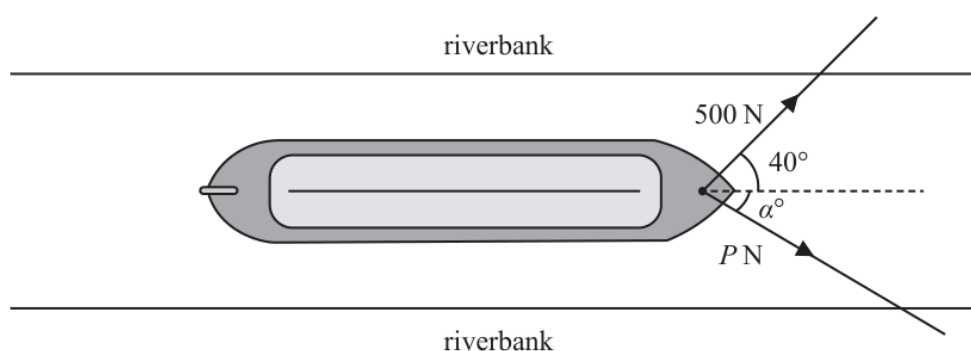


Figure 3

A boat is pulled along a river at a constant speed by two ropes.

The banks of the river are parallel and the boat travels horizontally in a straight line, parallel to the riverbanks.

- The tension in the first rope is 500 N acting at an angle of  $40^\circ$  to the direction of motion, as shown in Figure 3.
- The tension in the second rope is  $P$  newtons, acting at an angle of  $\alpha^\circ$  to the direction of motion, also shown in Figure 3.
- The resistance to motion of the boat as it moves through the water is a constant force of magnitude 900 N

The boat is modelled as a particle. The ropes are modelled as being light and lying in a horizontal plane.

Use the model to find

- the value of  $\alpha$
- the value of  $P$

(8)

**TOPIC**

# **Kinematics Graphs**



## Question 6

## Kinematics Graphs

6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time  $t = 0$  seconds and moves with uniform acceleration  $0.8 \text{ m s}^{-2}$  for  $T$  seconds, reaching a speed of  $V \text{ m s}^{-1}$ . She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (a) Sketch a speed-time graph for the model of the motion of the athlete from the instant when she starts to the instant when she crosses the finishing line. (2)
- (b) Write down an expression for  $V$  in terms of  $T$ . (1)
- (c) Show that  $T^2 - kT + 500 = 0$ , where  $k$  is a constant to be found. (4)
- (d) Hence find the value of  $T$ , justifying your answer carefully. (3)
- (e) Considering your speed-time graph or otherwise, state two ways, apart from including air resistance, in which the model could be made to be more realistic. (2)

## Question 7

## Kinematics Graphs

7. A helicopter is hovering at rest above horizontal ground at the point  $H$ . A parachutist steps out of the helicopter and immediately falls vertically and freely under gravity from rest for 2.5 s. His parachute then opens and causes him to immediately decelerate at a constant rate of  $3.9 \text{ m s}^{-2}$  for  $T$  seconds ( $T < 6$ ), until his speed is reduced to  $V \text{ m s}^{-1}$ . He then moves with this constant speed  $V \text{ m s}^{-1}$  until he hits the ground. While he is decelerating, he falls a distance of 73.75 m. The total time between the instant when he leaves  $H$  and the instant when he hits the ground is 20 s.

The parachutist is modelled as a particle.

- (a) Find the speed of the parachutist at the instant when his parachute opens. (1)
- (b) Sketch a speed-time graph for the motion of the parachutist from the instant when he leaves  $H$  to the instant when he hits the ground. (2)
- (c) Find the value of  $T$ . (5)
- (d) Find, to the nearest metre, the height of the point  $H$  above the ground. (4)

## Question 8

## Kinematics Graphs

8. Two trams, tram  $A$  and tram  $B$ , run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time  $t = 0$ , tram  $A$  starts to move. Tram  $A$  moves with constant acceleration  $2 \text{ m s}^{-2}$  for 5 seconds and then continues to move along the track at constant speed.

At time  $t = 20$  seconds, tram  $B$  starts from rest and moves in the same direction as tram  $A$ . Tram  $B$  moves with constant acceleration  $3 \text{ m s}^{-2}$  for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

- (a) Sketch, on the same axes, a speed-time graph for the motion of tram  $A$  and a speed-time graph for the motion of tram  $B$ , from  $t = 0$  to the instant when tram  $B$  overtakes tram  $A$ .

(3)

At the instant when the two trams are moving with the same speed, tram  $A$  is  $d$  metres in front of tram  $B$ .

- (b) Find the value of  $d$ .

(5)

- (c) Find the distance of the trams from the depot at the instant when tram  $B$  overtakes tram  $A$ .

(5)

---

(Total 13 marks)

## Question 7

## Kinematics Graphs

7. Two small children, Ajaz and Beth, are running a 100 m race along a straight horizontal track.

They both start from rest, leaving the start line at the same time.

Ajaz accelerates at  $0.8 \text{ ms}^{-2}$  up to a speed of  $4 \text{ ms}^{-1}$  and then maintains this speed until he crosses the finish line.

Beth accelerates at  $1 \text{ ms}^{-2}$  for  $T$  seconds and then maintains a constant speed until she crosses the finish line.

Ajaz and Beth cross the finish line at the same time.

- (a) Sketch, on the same axes, a speed-time graph for each child, from the instant when they leave the start line to the instant when they cross the finish line. (3)
- (b) Find the time taken by Ajaz to complete the race. (4)
- (c) Find the value of  $T$  (4)
- (d) Find the difference in the speeds of the two children as they cross the finish line. (2)

## Question 6

## Kinematics Graphs

6.

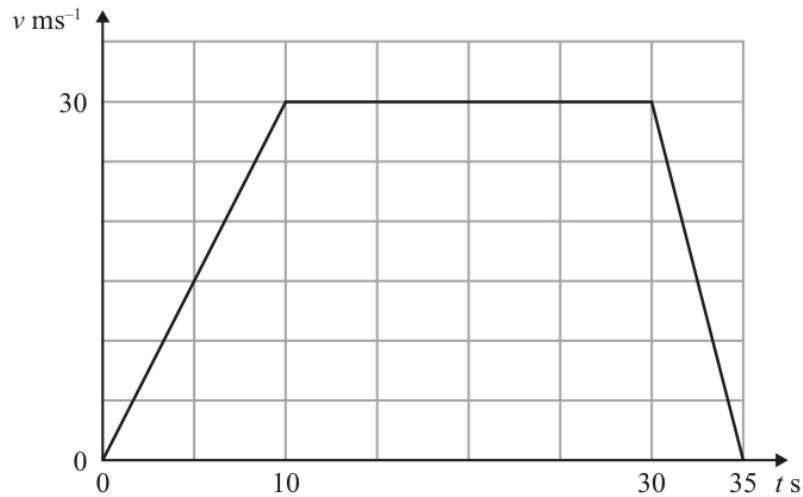


Figure 4

The point  $O$  lies on a straight horizontal road.

At time  $t = 0$ , a car leaves  $O$  and travels along the road.

The velocity-time graph in Figure 4 shows the velocity,  $v \text{ ms}^{-1}$ , of the car at time  $t$  seconds for the first 35 seconds of its journey.

(a) Find

(i) the acceleration of the car for the period  $0 \leq t \leq 10$

(ii) the deceleration of the car for the period  $30 \leq t \leq 35$

(2)

(b) Sketch an acceleration-time graph for the car for the period  $0 \leq t \leq 35$

(2)

(c) Find the distance travelled by the car for the period  $0 \leq t \leq 35$

(2)

When  $t = 5$ , a motorcycle starts from rest at  $O$ .

The motorcycle travels along the same road as the car and in the same direction.

For the period  $5 \leq t \leq 20$ , the acceleration of the motorcycle is  $A \text{ ms}^{-2}$ , where  $A$  is a positive constant.

The motorcycle catches up with the car when  $t = 20$

(d) Find the value of  $A$ .

(4)

14 marks

WME01/01 OCTOBER 2019

Kinematics Graphs

## Question 8

Also in Kinematics Graphs

Primary: Working with Vectors

8. [In this question, the horizontal unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

Two speedboats,  $A$  and  $B$ , are each moving with constant velocity. The velocity of  $A$  is  $20 \text{ km h}^{-1}$  due west and the velocity of  $B$  is  $40 \text{ km h}^{-1}$  on a bearing of  $150^\circ$ . The boats are modelled as particles.

At noon, the position vector of  $A$  is  $60\mathbf{i} \text{ km}$  and  $B$  is at the origin  $O$ . At time  $t$  hours after noon, the position vector of  $A$  is  $\mathbf{r} \text{ km}$  and the position vector of  $B$  is  $\mathbf{s} \text{ km}$ .

- (a) Find the velocity of  $B$  in the form  $(p\mathbf{i} + q\mathbf{j}) \text{ km h}^{-1}$  (3)
- (b) Find expressions for  $\mathbf{r}$  and  $\mathbf{s}$  in terms of  $t$ ,  $\mathbf{i}$  and  $\mathbf{j}$ . (3)
- (c) Find the time, to the nearest minute, at which the distance between the boats is the same as it was at noon. (8)

5 marks

WME01/01 OCTOBER 2021

Kinematics Graphs

## Question 6

Also in Kinematics Graphs

Primary: Constant Acceleration in 1D

6.

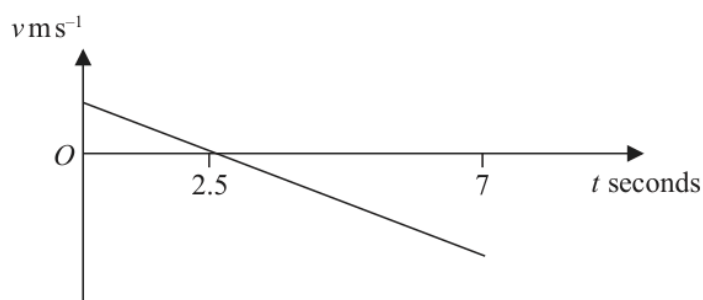


Figure 3

A small ball is thrown vertically upwards at time  $t = 0$  from a point  $A$  which is above horizontal ground. The ball hits the ground  $7 \text{ s}$  later.

The ball is modelled as a particle moving freely under gravity.

The velocity-time graph shown in Figure 3 represents the motion of the ball for  $0 \leq t \leq 7$

(a) Find the speed with which the ball is thrown. (2)

(b) Find the height of  $A$  above the ground. (3)

12 marks

WME01/01 JANUARY 2024

Kinematics Graphs

## Question 6

Also in Kinematics Graphs

Primary: Constant Acceleration in 1D

6. A particle is projected vertically upwards from a point  $A$  with speed  $24 \text{ m s}^{-1}$

The point  $A$  is  $2.5 \text{ m}$  vertically above the point  $B$ .

Point  $B$  lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at  $B$  with speed  $V \text{ m s}^{-1}$

After hitting the ground the particle does not rebound.

- (a) Find the value of  $V$ .

(3)

- (b) Find the time taken for the particle to reach  $B$ .

(3)

The point  $C$  is  $10 \text{ m}$  vertically above  $A$ .

- (c) Find the length of time for which the particle is above  $C$ .

(4)

- (d) Sketch a speed-time graph for the motion of the particle from projection to the instant that it reaches  $B$ . (No further calculations are required.)

(2)



TOPIC

# Constant Acceleration in 1D

## Question 6

## Constant Acceleration in 1D

6.

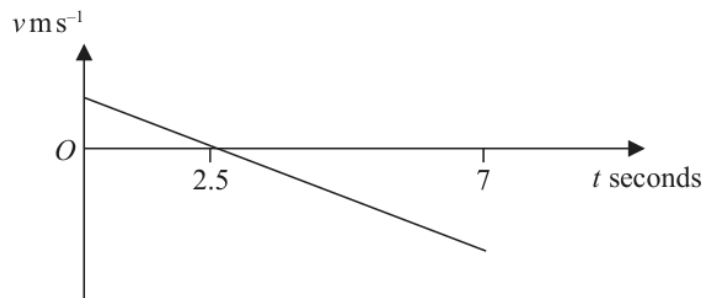


Figure 3

A small ball is thrown vertically upwards at time  $t = 0$  from a point  $A$  which is above horizontal ground. The ball hits the ground  $7 \text{ s}$  later.

The ball is modelled as a particle moving freely under gravity.

The velocity-time graph shown in Figure 3 represents the motion of the ball for  $0 \leq t \leq 7$

(a) Find the speed with which the ball is thrown. (2)

(b) Find the height of  $A$  above the ground. (3)

## Question 6

## Constant Acceleration in 1D

6. A particle is projected vertically upwards from a point  $A$  with speed  $24\text{ m s}^{-1}$

The point  $A$  is  $2.5\text{ m}$  vertically above the point  $B$ .

Point  $B$  lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at  $B$  with speed  $V\text{ m s}^{-1}$

After hitting the ground the particle does not rebound.

- (a) Find the value of  $V$ .

(3)

- (b) Find the time taken for the particle to reach  $B$ .

(3)

The point  $C$  is  $10\text{ m}$  vertically above  $A$ .

- (c) Find the length of time for which the particle is above  $C$ .

(4)

- (d) Sketch a speed-time graph for the motion of the particle from projection to the instant that it reaches  $B$ . (No further calculations are required.)

(2)

## Question 7

## Constant Acceleration in 1D

- 7 At time  $t = 0$ , a small ball  $A$  is projected vertically upwards with speed  $8 \text{ m s}^{-1}$  from a fixed point on horizontal ground.

The ball hits the ground again for the first time at time  $t = T_1$  seconds.

Ball  $A$  is modelled as a particle moving freely under gravity.

- (a) Show that  $T_1 = 1.63$  to 3 significant figures.

(2)

After the first impact with the ground,  $A$  rebounds to a height of 2 m above the ground.

Given that the mass of  $A$  is 0.1 kg,

- (b) find the magnitude of the impulse received by  $A$  as a result of its first impact with the ground.

(5)

At time  $t = 1$  second, another small ball  $B$  is projected vertically upwards from another point on the ground with speed  $5 \text{ m s}^{-1}$

Ball  $B$  is modelled as a particle moving freely under gravity.

At time  $t = T_2$  seconds ( $T_2 > 1$ ),  $A$  and  $B$  are at the same height above the ground for the first time.

- (c) Find the value of  $T_2$

(4)

**18 marks**

**WME01/01 JANUARY 2020**

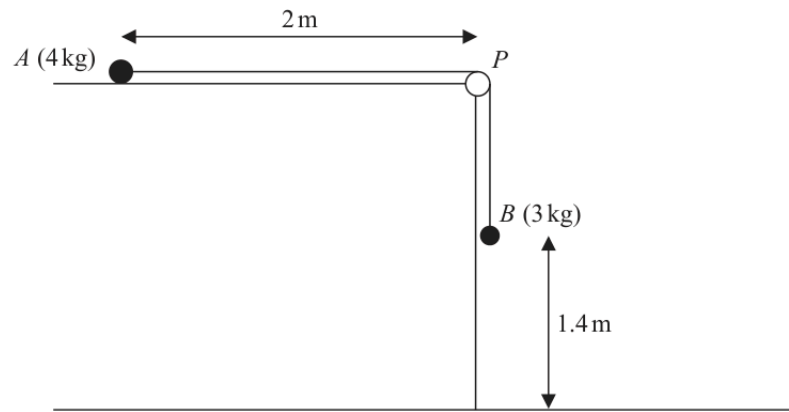
## **Question 7**

**Constant Acceleration in 1D**

**Also in Constant Acceleration in 1D**

**Primary: Newton's Second Law**

7.



**Figure 3**

A particle  $A$  of mass  $4\text{ kg}$  is held at rest on a rough horizontal table. Particle  $A$  is attached to one end of a string that passes over a pulley  $P$ . The pulley is fixed at the edge of the table. The other end of the string is attached to a particle  $B$ , of mass  $3\text{ kg}$ , which hangs freely below  $P$ .

The part of the string from  $A$  to  $P$  is perpendicular to the edge of the table and  $A$ ,  $P$  and  $B$  all lie in the same vertical plane.

The string is modelled as being light and inextensible and the pulley is modelled as being small, smooth and light.

The system is released from rest with the string taut. At the instant of release,  $A$  is  $2\text{ m}$  from the edge of the table and  $B$  is  $1.4\text{ m}$  above a horizontal floor, as shown in Figure 3.

After descending with constant acceleration for  $2\text{ seconds}$ ,  $B$  hits the floor and does not rebound.

(a) Show that the acceleration of  $A$  before  $B$  hits the floor is  $0.7\text{ ms}^{-2}$  (2)

(b) State which of the modelling assumptions you have used in order to answer part (a). (1)

(c) Find the magnitude of the resultant force exerted on the pulley by the string. (4)

The coefficient of friction between  $A$  and the table is  $\mu$ .

(d) Find the value of  $\mu$ . (6)

(e) Determine, by calculation, whether or not  $A$  reaches the pulley. (5)

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**(Total 18 marks)**

12 marks

WME01/01 JANUARY 2021

## Question 7

Constant Acceleration in 1D

Also in Constant Acceleration in 1D

Primary: Kinematics Graphs

7. A helicopter is hovering at rest above horizontal ground at the point  $H$ . A parachutist steps out of the helicopter and immediately falls vertically and freely under gravity from rest for 2.5 s. His parachute then opens and causes him to immediately decelerate at a constant rate of  $3.9 \text{ m s}^{-2}$  for  $T$  seconds ( $T < 6$ ), until his speed is reduced to  $V \text{ m s}^{-1}$ . He then moves with this constant speed  $V \text{ m s}^{-1}$  until he hits the ground. While he is decelerating, he falls a distance of 73.75 m. The total time between the instant when he leaves  $H$  and the instant when he hits the ground is 20 s.

The parachutist is modelled as a particle.

- (a) Find the speed of the parachutist at the instant when his parachute opens. (1)
- (b) Sketch a speed-time graph for the motion of the parachutist from the instant when he leaves  $H$  to the instant when he hits the ground. (2)
- (c) Find the value of  $T$ . (5)
- (d) Find, to the nearest metre, the height of the point  $H$  above the ground. (4)

17 marks

WME01/01 JANUARY 2021

Constant Acceleration in 1D

## Question 8

Also in Constant Acceleration in 1D

Primary: Resolving Forces, Inclined Planes

8.

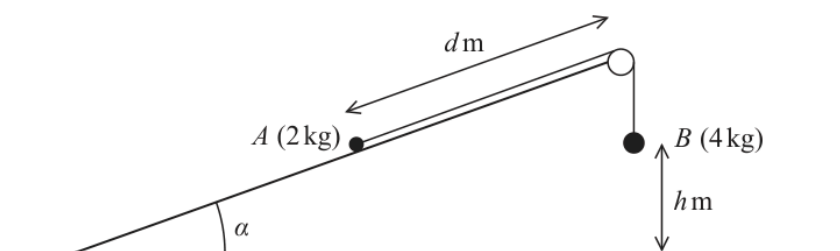


Figure 3

Two particles,  $A$  and  $B$ , have masses  $2\text{ kg}$  and  $4\text{ kg}$  respectively. The particles are connected by a light inextensible string. The string passes over a small smooth pulley which is fixed at the top of a rough plane. The plane is inclined to the horizontal ground at an angle  $\alpha$  where  $\tan \alpha = \frac{3}{4}$ . The particle  $A$  is held at rest on the plane at a distance  $d$  metres from the pulley. The particle  $B$  hangs freely at rest, vertically below the pulley, at a distance  $h$  metres above the ground, as shown in Figure 3. The part of the string between  $A$  and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between  $A$  and the plane is  $\frac{1}{4}$ .

The system is released from rest with the string taut and  $B$  descends.

- (a) Find the tension in the string as  $B$  descends.

(9)

On hitting the ground,  $B$  immediately comes to rest.

Given that  $A$  comes to rest before reaching the pulley,

- (b) find, in terms of  $h$ , the range of possible values of  $d$ .

(7)

- (c) State one physical factor, other than air resistance, that could be taken into account to make the model described above more realistic.

(1)

(Total 17 marks)



13 marks

WME01/01 MAY/JUNE  
2021

Constant Acceleration in 1D

## Question 6

Also in Constant Acceleration in 1D

Primary: Resolving Forces, Inclined Planes

6. A fixed rough plane is inclined at an angle  $\theta$  to the horizontal, where  $\tan \theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed  $5 \text{ m s}^{-1}$  from a point  $A$  on the plane, up a line of greatest slope of the plane.

The coefficient of friction between the particle and the plane is  $\frac{1}{4}$

- (a) Find the magnitude of the frictional force acting on the particle as it moves up the plane. (3)

The particle comes to instantaneous rest at the point  $B$ .

- (b) Find the distance  $AB$ . (5)

The particle now slides down the plane from  $B$ . At the instant when the particle passes through the point  $C$  on the plane, the speed of the particle is again  $5 \text{ m s}^{-1}$

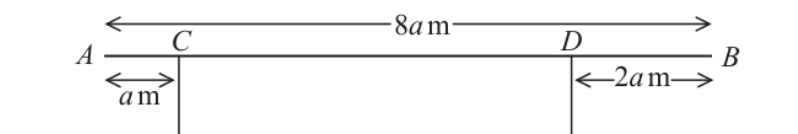
- (c) Find the distance  $BC$ . (5)

**10 marks****WME01/01 MAY/JUNE 2021****Constant Acceleration in 1D**

## Question 7

**Also in Constant Acceleration in 1D****Primary: Moments**

7.

**Figure 2**

A non-uniform beam  $AB$ , of mass  $60\text{ kg}$  and length  $8a$  metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at  $C$ , where  $AC = a$  metres and the other support is at  $D$ , where  $DB = 2a$  metres, as shown in Figure 2.

The magnitude of the normal reaction between the beam and the support at  $D$  is three times the magnitude of the normal reaction between the beam and the support at  $C$ .

By modelling the beam as a non-uniform rod whose centre of mass is at a distance  $x$  metres from  $A$ ,

- (a) find an expression for  $x$  in terms of  $a$ .

**(5)**

A box of mass  $M$  kg is placed on the beam at  $E$ , where  $AE = 2a$  metres.

The beam remains in equilibrium in a horizontal position.

The magnitude of the normal reaction between the beam and the support at  $C$  is now equal to the magnitude of the normal reaction between the beam and the support at  $D$ .

By modelling the box as a particle,

- (b) find the value of  $M$ .

**(5)**

13 marks

WME01/01 MAY/JUNE 2021

## Question 8

Constant Acceleration in 1D

Also in Constant Acceleration in 1D

Primary: Kinematics Graphs

8. Two trams, tram  $A$  and tram  $B$ , run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time  $t = 0$ , tram  $A$  starts to move. Tram  $A$  moves with constant acceleration  $2 \text{ m s}^{-2}$  for 5 seconds and then continues to move along the track at constant speed.

At time  $t = 20$  seconds, tram  $B$  starts from rest and moves in the same direction as tram  $A$ . Tram  $B$  moves with constant acceleration  $3 \text{ m s}^{-2}$  for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

- (a) Sketch, on the same axes, a speed-time graph for the motion of tram  $A$  and a speed-time graph for the motion of tram  $B$ , from  $t = 0$  to the instant when tram  $B$  overtakes tram  $A$ .

(3)

At the instant when the two trams are moving with the same speed, tram  $A$  is  $d$  metres in front of tram  $B$ .

- (b) Find the value of  $d$ .

(5)

- (c) Find the distance of the trams from the depot at the instant when tram  $B$  overtakes tram  $A$ .

(5)

**(Total 13 marks)**

---

13 marks

WME01/01 MAY/JUNE 2022

## Question 7

Constant Acceleration in 1D

Also in Constant Acceleration in 1D

Primary: Kinematics Graphs

7. Two small children, Ajaz and Beth, are running a 100 m race along a straight horizontal track.

They both start from rest, leaving the start line at the same time.

Ajaz accelerates at  $0.8 \text{ ms}^{-2}$  up to a speed of  $4 \text{ ms}^{-1}$  and then maintains this speed until he crosses the finish line.

Beth accelerates at  $1 \text{ ms}^{-2}$  for  $T$  seconds and then maintains a constant speed until she crosses the finish line.

Ajaz and Beth cross the finish line at the same time.

- (a) Sketch, on the same axes, a speed-time graph for each child, from the instant when they leave the start line to the instant when they cross the finish line.

(3)

- (b) Find the time taken by Ajaz to complete the race.

(4)

- (c) Find the value of  $T$

(4)

- (d) Find the difference in the speeds of the two children as they cross the finish line.

(2)

7 marks

WME01/01 JANUARY 2023

## Question 7

Constant Acceleration in 1D

Also in Constant Acceleration in 1D

Primary: Newton's Second Law

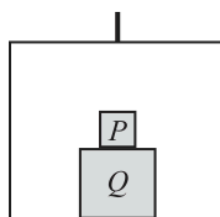


Figure 4

A simple lift operates by means of a vertical cable which is attached to the top of the lift.

The lift has mass  $m$

A box  $Q$  is placed on the floor of the lift.

A box  $P$  is placed directly on top of box  $Q$ , as shown in Figure 4.

The cable is modelled as being light and inextensible and air resistance is modelled as being negligible.

The tension in the cable is  $\frac{42mg}{5}$

The lift and its contents move vertically upwards with acceleration  $\frac{2g}{5}$

Using the model,

(a) find, in terms of  $m$ , the combined mass of boxes  $P$  and  $Q$

(4)

During the motion of the lift, the force exerted on box  $P$  by box  $Q$  is  $\frac{14mg}{5}$

Using the model,

(b) find, in terms of  $m$ , the mass of box  $P$

(3)

15 marks

WME01/01 JANUARY  
2023

Constant Acceleration in 1D

## Question 8

Also in Constant Acceleration in 1D

Primary: Resolving Forces, Inclined Planes

8.

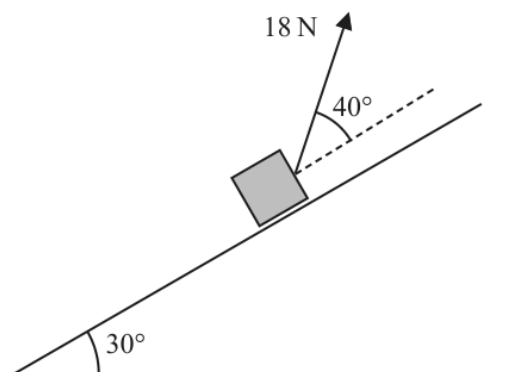


Figure 5

A parcel of mass 2 kg is pulled up a rough inclined plane by the action of a constant force.

The force has magnitude 18 N and acts at an angle of  $40^\circ$  to the plane.

The line of action of the force lies in a vertical plane containing a line of greatest slope of the inclined plane.

The plane is inclined at an angle of  $30^\circ$  to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is 0.3

The parcel is modelled as a particle  $P$

(a) Find the acceleration of  $P$

(8)

The points  $A$  and  $B$  lie on a line of greatest slope of the plane, where  $AB = 5$  m and  $B$  is above  $A$ . Particle  $P$  passes through  $A$  with speed  $2 \text{ m s}^{-1}$  in the direction  $AB$ .

(b) Find the speed of  $P$  as it passes through  $B$ .

(3)

The force of 18 N is removed at the instant  $P$  passes through  $B$ . As a result,  $P$  comes to rest at the point  $C$ .

(c) Determine whether  $P$  will remain at rest at  $C$ . You must show all stages of your working clearly.

(4)

11 marks

WME01/01 MAY/JUNE 2023

## Question 7

Constant Acceleration in 1D

Also in Constant Acceleration in 1D

Primary: Newton's Second Law

7.

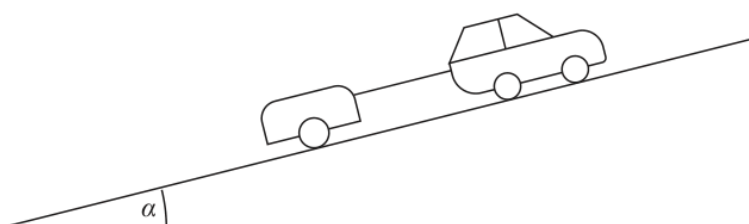


Figure 4

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle  $\alpha$  to the horizontal, where  $\sin \alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration  $0.75 \text{ m s}^{-2}$

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude  $2R$  newtons
- the **trailer** is modelled as a constant force of magnitude  $R$  newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of  $R$  is 60

(4)

(b) find the tension in the tow bar.

(3)

When the car and trailer are moving at a speed of  $12 \text{ m s}^{-1}$ , the tow bar breaks.

Given that the non-gravitational resistance to motion of the trailer remains unchanged,

(c) use the model to find the further distance moved by the trailer before it first comes to rest.

(4)

12 marks

WME01/01 JANUARY  
2024

Constant Acceleration in 1D

## Question 8

Also in Constant Acceleration in 1D

Primary: Resolving Forces, Inclined Planes

8.

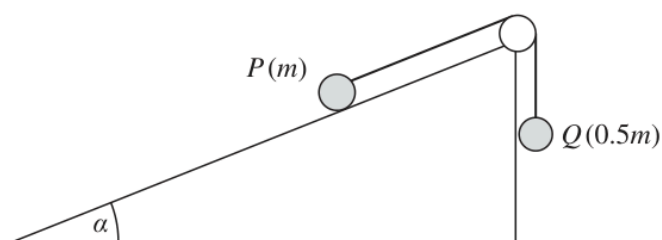


Figure 5

A fixed rough plane is inclined at an angle  $\alpha$  to the horizontal, where  $\tan \alpha = \frac{5}{12}$

A small smooth pulley is fixed at the top of the plane.

One end of a light inextensible string is attached to a particle  $P$  which is at rest on the plane. The string passes over the pulley and the other end of the string is attached to a particle  $Q$  which hangs vertically below the pulley, as shown in Figure 5.

Particle  $P$  has mass  $m$  and particle  $Q$  has mass  $0.5m$

The string from  $P$  to the pulley lies along a line of greatest slope of the plane.

The coefficient of friction between  $P$  and the plane is  $\mu$ .

The system is in **limiting equilibrium** with the string taut and  $P$  is on the point of slipping **up** the plane.

(a) Find the value of  $\mu$ .

(8)

The string breaks and  $P$  begins to move down the plane.

When particle  $P$  has travelled a distance of 0.8 m down the plane, the speed of  $P$  is  $V \text{ m s}^{-1}$

(b) Find the value of  $V$ .

(4)



12 marks

WME01/01 MAY/JUNE  
2024

Constant Acceleration in 1D

## Question 6

Also in Constant Acceleration in 1D

Primary: Resolving Forces, Inclined Planes

6.

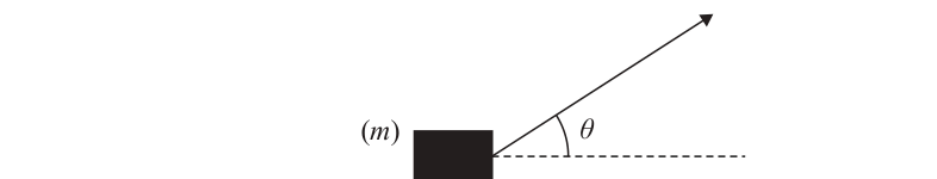


Figure 3

A box of mass  $m$  lies on a rough horizontal plane. The box is pulled along the plane in a straight line at **constant speed** by a light rope. The rope is inclined at an angle  $\theta$  to the plane, as shown in Figure 3.

The coefficient of friction between the box and the plane is  $\frac{1}{3}$

The box is modelled as a particle.

Given that  $\tan \theta = \frac{3}{4}$

(a) find, in terms of  $m$  and  $g$ , the tension in the rope.

(7)

The rope is now removed and the box is placed at rest on the plane.

The box is then projected horizontally along the plane with speed  $u$ .

The box is again modelled as a particle.

When the box has moved a distance  $d$  along the plane, the speed of the box is  $\frac{1}{2}u$ .

(b) Find  $d$  in terms of  $u$  and  $g$ .

(5)

12 marks

WME01/01 OCTOBER 2024

Constant Acceleration in 1D

## Question 6

Also in Constant Acceleration in 1D

Primary: Resolving Forces, Inclined Planes

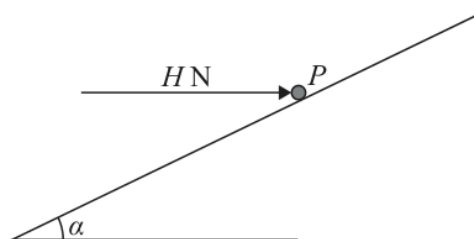


Figure 4

A particle  $P$  of mass 5 kg lies on the surface of a rough plane.

The plane is inclined at an angle  $\alpha$  to the horizontal, where  $\tan \alpha = \frac{3}{4}$

The particle is held in equilibrium by a horizontal force of magnitude  $H$  newtons, as shown in Figure 4.

The horizontal force acts in a vertical plane containing a line of greatest slope of the inclined plane.

The coefficient of friction between the particle and the plane is  $\frac{1}{4}$

(a) Find the smallest possible value of  $H$ .

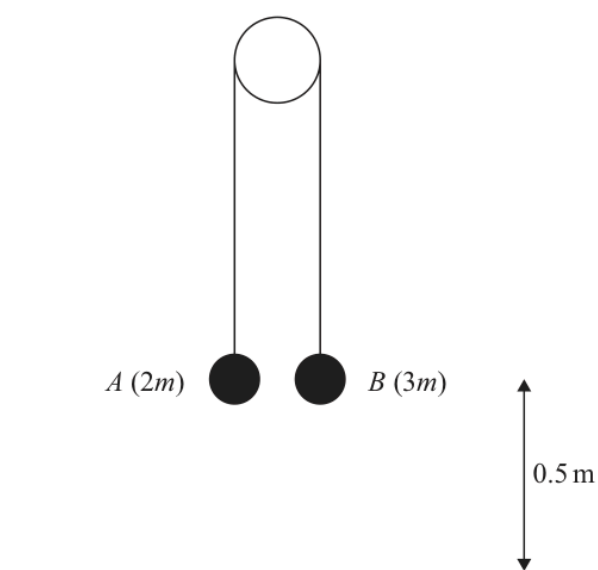
(6)

The horizontal force is now removed, and  $P$  starts to slide down the slope.

In the first  $T$  seconds after  $P$  is released from rest,  $P$  slides 1.5 m down the slope.

(b) Find the value of  $T$ .

(6)

**15 marks****WME01/01 JANUARY 2025****Question 7****Constant Acceleration in 1D****Also in Constant Acceleration in 1D****Primary: Newton's Second Law****7.****Figure 4**

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end is attached to a particle  $B$  of mass  $3m$ . The string passes over a small smooth fixed pulley. The string is taut and both straight parts of the string are vertical. Both particles are held at a distance  $0.5\text{ m}$  above a horizontal surface, as shown in Figure 4.

The system is released from rest and  $B$  moves downwards.

(a) Find the tension in the string in terms of  $m$  and  $g$ . (5)

(b) Find the speed of  $B$  at the instant it strikes the surface. (4)

In the subsequent motion,  $A$  does not reach the pulley and  $B$  does not rebound after it strikes the surface.

(c) Find the time from the instant when the system is released from rest to the instant when  $A$  first reaches a height of  $1.06\text{ m}$  above the surface. (6)

11 marks

WME01/01 MAY/JUNE  
2025

Constant Acceleration in 1D

## Question 6

Also in Constant Acceleration in 1D

Primary: Resolving Forces, Inclined Planes

6.

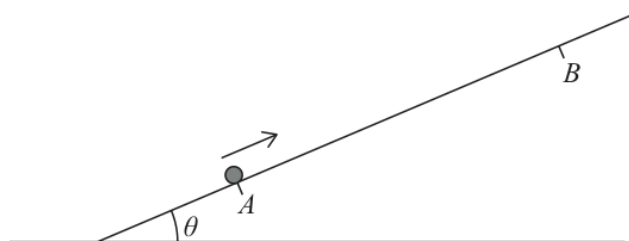


Figure 4

Figure 4 shows a rough plane inclined at an angle  $\theta$  to the horizontal, where  $\tan \theta = \frac{3}{4}$ .

The points  $A$  and  $B$  lie on a line of greatest slope of the plane, with  $B$  above  $A$ .

A package  $P$  of mass  $m$  is projected up the plane from  $A$  towards  $B$ .

The coefficient of friction between the plane and the package is  $\frac{1}{4}$ .

The package  $P$  is modelled as a particle.

- (a) Show that the **deceleration** of  $P$ , as it moves from  $A$  to  $B$ , is  $\frac{4}{5}g$  (6)

The package  $P$  comes to rest at  $B$ .

Given that  $P$  is projected from  $A$  with speed  $U \text{ m s}^{-1}$  and that  $AB = 1.5 \text{ m}$ ,

- (b) find the value of  $U$ . (2)

On reaching  $B$ ,  $P$  is held at rest there by a force of magnitude  $X$  newtons acting up the plane in the direction  $AB$ .

Given that the mass of  $P$  is  $2 \text{ kg}$ ,

- (c) find the smallest possible value of  $X$ . (3)

**10 marks**

**WME01/01 OCTOBER 2025**

## **Question 6**

**Constant Acceleration in 1D**

**Also in Constant Acceleration in 1D**

**Primary: Kinematics Graphs**

6.

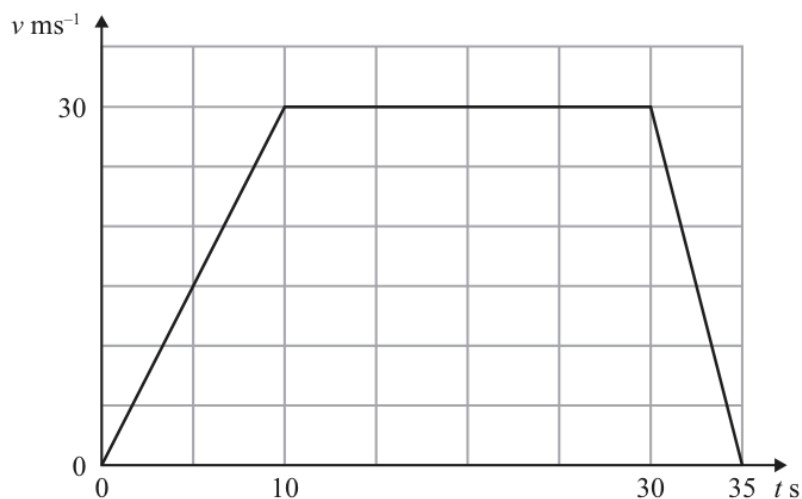


Figure 4

The point  $O$  lies on a straight horizontal road.

At time  $t = 0$ , a car leaves  $O$  and travels along the road.

The velocity-time graph in Figure 4 shows the velocity,  $v \text{ m s}^{-1}$ , of the car at time  $t$  seconds for the first 35 seconds of its journey.

(a) Find

(i) the acceleration of the car for the period  $0 \leq t \leq 10$

(ii) the deceleration of the car for the period  $30 \leq t \leq 35$

(2)

(b) Sketch an acceleration-time graph for the car for the period  $0 \leq t \leq 35$

(2)

(c) Find the distance travelled by the car for the period  $0 \leq t \leq 35$

(2)

When  $t = 5$ , a motorcycle starts from rest at  $O$ .

The motorcycle travels along the same road as the car and in the same direction.

For the period  $5 \leq t \leq 20$ , the acceleration of the motorcycle is  $A \text{ m s}^{-2}$ , where  $A$  is a positive constant.

The motorcycle catches up with the car when  $t = 20$

(d) Find the value of  $A$ .

(4)

TOPIC

# Constant Acceleration in 2D

12 marks

WME01/01 JANUARY 2021

## Question 6

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

At time  $t = 0$ , Agatha is at the point with position vector  $(11\mathbf{i} + 11\mathbf{j})\text{m}$  and Brionie is at the point with position vector  $(7\mathbf{i} + 16\mathbf{j})\text{m}$ . The position vectors are given relative to the door,  $O$ , and  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal perpendicular unit vectors.

Agatha skates with constant velocity  $(3\mathbf{i} - \mathbf{j})\text{ms}^{-1}$

Brionie skates with constant velocity  $(4\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$

- (a) Find the position vector of Agatha at time  $t$  seconds. (2)

At time  $t = 6$  seconds, Agatha passes through the point  $P$ .

- (b) Show that Brionie also passes through  $P$  and find the value of  $t$  when this occurs. (4)

At time  $t$  seconds, Agatha is at the point  $A$  and Brionie is at the point  $B$ .

- (c) Show that  $\overrightarrow{AB} = [(t - 4)\mathbf{i} + (5 - t)\mathbf{j}]\text{m}$  (2)

- (d) Find the distance between the two girls when they are closest together. (4)



12 marks

WME01/01 OCTOBER 2021

## Question 8

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

8. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

At 7 am a ship leaves a port and moves with constant velocity. The position vector of the port is  $(-2\mathbf{i} + 9\mathbf{j})$  km.

At 7.36 am the ship is at the point with position vector  $(4\mathbf{i} + 6\mathbf{j})$  km.

- (a) Show that the velocity of the ship is  $(10\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$  (2)

- (b) Find the position vector of the ship  $t$  hours after leaving port. (2)

At 8.48 am a passenger on the ship notices that a lighthouse is due east of the ship.

At 9 am the same passenger notices that the lighthouse is now north east of the ship.

- (c) Find the position vector of the lighthouse. (4)

- (d) Find the position vector of the ship when it is due south of the lighthouse. (4)

---

(Total 12 marks)

12 marks

WME01/01 JANUARY 2022

## Question 6

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

6. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors.]

A particle  $P$  of mass  $2\text{ kg}$  moves under the action of two forces,  $(p\mathbf{i} + q\mathbf{j})\text{N}$  and  $(2q\mathbf{i} + p\mathbf{j})\text{N}$ , where  $p$  and  $q$  are constants.

Given that the acceleration of  $P$  is  $(\mathbf{i} - \mathbf{j})\text{ms}^{-2}$

(a) find the value of  $p$  and the value of  $q$ .

(5)

(b) Find the size of the angle between the direction of the acceleration and the vector  $\mathbf{j}$ .

(2)

At time  $t = 0$ , the velocity of  $P$  is  $(3\mathbf{i} - 4\mathbf{j})\text{ms}^{-1}$

At  $t = T$  seconds,  $P$  is moving in the direction of the vector  $(11\mathbf{i} - 13\mathbf{j})$ .

(c) Find the value of  $T$ .

(5)

14 marks

WME01/01 JANUARY 2022

## Question 8

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

8. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

A ship  $A$  moves with constant velocity  $(3\mathbf{i} - 10\mathbf{j}) \text{ km h}^{-1}$

At time  $t$  hours, the position vector of  $A$  is  $\mathbf{r}$  km.

At time  $t = 0$ ,  $A$  is at the point with position vector  $(13\mathbf{i} + 5\mathbf{j}) \text{ km}$ .

- (a) Find  $\mathbf{r}$  in terms of  $t$ .

(2)

Another ship  $B$  moves with constant velocity  $(15\mathbf{i} + 14\mathbf{j}) \text{ km h}^{-1}$

At time  $t = 0$ ,  $B$  is at the point with position vector  $(3\mathbf{i} - 5\mathbf{j}) \text{ km}$ .

- (b) Show that, at time  $t$  hours,

$$\overrightarrow{AB} = [(12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}] \text{ km}$$

(4)

Given that the two ships do not change course,

- (c) find the shortest distance between the two ships,

(6)

- (d) find the bearing of ship  $B$  from ship  $A$  when the ships are closest.

(2)

(Total 14 marks)

6 marks

WME01/01 MAY/JUNE 2022

## Question 6

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

6. A particle  $P$  is moving with constant acceleration.

At time  $t = 1$  second,  $P$  has velocity  $(-\mathbf{i} + 4\mathbf{j})\text{ms}^{-1}$

At time  $t = 4$  seconds,  $P$  has velocity  $(5\mathbf{i} - 8\mathbf{j})\text{ms}^{-1}$

Find the speed of  $P$  at time  $t = 3.5$  seconds.

(6)

17 marks

WME01/01 MAY/JUNE 2022

## Question 8

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

8. [In this question,  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

Two boats,  $P$  and  $Q$ , are moving with constant velocities.

The velocity of  $P$  is  $15\mathbf{i}\text{ m s}^{-1}$  and the velocity of  $Q$  is  $(20\mathbf{i} - 20\mathbf{j})\text{ m s}^{-1}$

- (a) Find the direction in which  $Q$  is travelling, giving your answer as a bearing. (2)

The boats are modelled as particles.

At time  $t = 0$ ,  $P$  is at the origin  $O$  and  $Q$  is at the point with position vector  $200\mathbf{j}\text{ m}$ .

At time  $t$  seconds, the position vector of  $P$  is  $\mathbf{p}\text{ m}$  and the position vector of  $Q$  is  $\mathbf{q}\text{ m}$ .

- (b) Show that

$$\overrightarrow{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}]\text{ m} \quad (5)$$

- (c) Find the bearing of  $P$  from  $Q$  when  $t = 10$  (2)

- (d) Find the distance between  $P$  and  $Q$  when  $Q$  is north east of  $P$  (5)

- (e) Find the times when  $P$  and  $Q$  are 200 m apart. (3)

(Total 17 marks)

16 marks

WME01/01 OCTOBER 2022

## Question 8

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

8. [In this question,  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

Two ships,  $A$  and  $B$ , are moving with constant velocities.

The velocity of  $A$  is  $(3\mathbf{i} + 12\mathbf{j}) \text{ km h}^{-1}$  and the velocity of  $B$  is  $(p\mathbf{i} + q\mathbf{j}) \text{ km h}^{-1}$

- (a) Find the speed of  $A$ .

(2)

The ships are modelled as particles.

At 12 noon,  $A$  is at the point with position vector  $(-9\mathbf{i} + 6\mathbf{j}) \text{ km}$  and  $B$  is at the point with position vector  $(16\mathbf{i} + 6\mathbf{j}) \text{ km}$ .

At time  $t$  hours after 12 noon,

$$\overrightarrow{AB} = [(25 - 12t)\mathbf{i} - 9t\mathbf{j}] \text{ km}$$

- (b) Find the value of  $p$  and the value of  $q$ .

(7)

- (c) Find the bearing of  $A$  from  $B$  when the ships are 15 km apart, giving your answer to the nearest degree.

(7)

(Total 16 marks)

**9 marks**

**WME01/01 MAY/JUNE 2023**

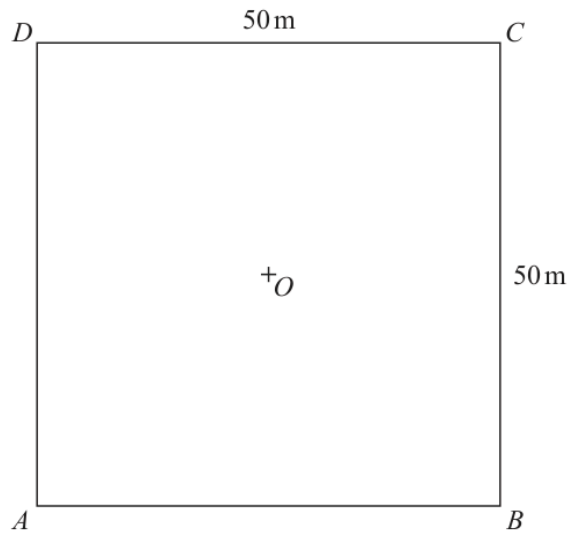
## **Question 8**

**Constant Acceleration in 2D**

**Also in Constant Acceleration in 2D**

**Primary: Working with Vectors**

8.



**Figure 5**

A square floor space  $ABCD$ , with centre  $O$ , is modelled as a flat horizontal surface measuring 50 m by 50 m, as shown in Figure 5.

The horizontal unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are in the direction of  $\overrightarrow{AB}$  and  $\overrightarrow{AD}$  respectively.

All position vectors are given relative to  $O$ .

A small robot  $R$  is programmed to travel across the floor at a constant velocity.

- At time  $t = 0$ ,  $R$  is at the point with position vector  $(-2\mathbf{i} + \mathbf{j})\text{m}$
- At time  $t = 11\text{ s}$ ,  $R$  is at the point with position vector  $(9\mathbf{i} + 23\mathbf{j})\text{m}$
- At time  $t$  seconds, the position vector of  $R$  is  $\mathbf{r}$  metres

(a) Find, in terms of  $t$ ,  $\mathbf{i}$  and  $\mathbf{j}$ , an expression for  $\mathbf{r}$

**(3)**

A second robot  $S$  is at the point  $C$ .

- At time  $t = 0$ ,  $S$  leaves  $C$  and moves with constant velocity  $(-\mathbf{i} - \mathbf{j})\text{m s}^{-1}$
- At time  $t$  seconds, the position vector of  $S$  is  $\mathbf{s}$  metres

(b) Write down, in terms of  $t$ ,  $\mathbf{i}$  and  $\mathbf{j}$ , an expression for  $\mathbf{s}$

**(1)**

(c) Show that

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}] \text{ m}$$

**(2)**

(d) Find the time when the distance between  $R$  and  $S$  is a minimum.

**(3)**



15 marks

WME01/01 OCTOBER 2023

## Question 6

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

6. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

At 12:00, a ship  $P$  sets sail from a harbour with position vector  $(15\mathbf{i} + 36\mathbf{j})$  km.

At 12:30,  $P$  is at the point with position vector  $(20\mathbf{i} + 34\mathbf{j})$  km.

Given that  $P$  moves with constant velocity,

- (a) show that the velocity of  $P$  is  $(10\mathbf{i} - 4\mathbf{j}) \text{ km h}^{-1}$  (2)

At time  $t$  hours after 12:00, the position vector of  $P$  is  $\mathbf{p}$  km.

- (b) Find an expression for  $\mathbf{p}$  in terms of  $\mathbf{i}$ ,  $\mathbf{j}$  and  $t$ . (2)

A second ship  $Q$  is also travelling at a constant velocity.

At time  $t$  hours after 12:00, the position vector of  $Q$  is given by  $\mathbf{q}$  km, where

$$\mathbf{q} = (42 - 8t)\mathbf{i} + (9 + 14t)\mathbf{j}$$

Ships  $P$  and  $Q$  are modelled as particles.

If both ships maintained their course,

- (c) (i) verify that they would collide at 13:30  
(ii) find the position vector of the point at which the collision would occur. (4)

At 12:30  $Q$  changes speed and direction to avoid the collision.

Ship  $Q$  now travels due north with a constant speed of  $15 \text{ km h}^{-1}$

Ship  $P$  maintains the same constant velocity throughout.

- (d) Find the exact distance between  $P$  and  $Q$  at 14:30 (7)

11 marks

WME01/01 JANUARY 2024

## Question 7

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

7. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

At midnight, a ship  $S$  is at the point with position vector  $(19\mathbf{i} + 22\mathbf{j})\text{ km}$

The ship travels with constant velocity  $(12\mathbf{i} - 16\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of  $S$ . (2)

At time  $t$  hours after midnight, the position vector of  $S$  is  $\mathbf{s}$  km.

- (b) Find an expression for  $\mathbf{s}$  in terms of  $\mathbf{i}$ ,  $\mathbf{j}$  and  $t$ . (2)

A lighthouse stands on a small rocky island. The lighthouse is modelled as being at the point with position vector  $(26\mathbf{i} + 15\mathbf{j})\text{ km}$ .

It is not safe for ships to be within 1.3 km of the lighthouse.

- (c) (i) Find the value of  $t$  when  $S$  is closest to the lighthouse.  
(ii) Hence determine whether it is safe for  $S$  to continue its course. (7)

13 marks

WME01/01 MAY/JUNE 2024

Constant Acceleration in 2D

## Question 7

Also in Constant Acceleration in 2D

Primary: Working with Vectors

7. [In this question, the horizontal unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

Two speedboats,  $A$  and  $B$ , are each moving with constant velocity.

- the velocity of  $A$  is  $40 \text{ km h}^{-1}$  due east
- the velocity of  $B$  is  $20 \text{ km h}^{-1}$  on a bearing of angle  $\alpha$  ( $0^\circ < \alpha < 90^\circ$ ), where  $\tan \alpha = \frac{4}{3}$

The boats are modelled as particles.

- (a) Find, in terms of  $\mathbf{i}$  and  $\mathbf{j}$ , the velocity of  $B$  in  $\text{km h}^{-1}$  (2)

At noon

- the position vector of  $A$  is  $20\mathbf{j}$  km
- the position vector of  $B$  is  $(10\mathbf{i} + 5\mathbf{j})$  km

At time  $t$  hours after noon

- the position vector of  $A$  is  $\mathbf{r}$  km, where  $\mathbf{r} = 20\mathbf{j} + 40t\mathbf{i}$
- the position vector of  $B$  is  $\mathbf{s}$  km

- (b) Find an expression for  $\mathbf{s}$  in terms of  $t$ ,  $\mathbf{i}$  and  $\mathbf{j}$ . (2)

- (c) Show that at time  $t$  hours after noon,

$$\overrightarrow{AB} = [(10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}] \text{ km} \quad (2)$$

- (d) Show that the boats will never collide. (3)

- (e) Find the distance between the boats when the bearing of  $B$  from  $A$  is  $225^\circ$  (4)

13 marks

WME01/01 MAY/JUNE 2025

## Question 7

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

7. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors.]

Two boats,  $A$  and  $B$ , are each moving with constant acceleration.

At time  $t$  hours after noon, boat  $A$  has velocity  $\mathbf{v}_A \text{ km h}^{-1}$ , where

$$\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$$

- (a) Find the magnitude of the acceleration of  $A$ . (2)

When  $t = 2$ , the velocity of  $B$  is  $(4\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

When  $t = 5$ , the velocity of  $B$  is  $(\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$

- (b) Find the acceleration of  $B$ , giving your answer in terms of  $\mathbf{i}$  and  $\mathbf{j}$ . (2)

- (c) Find the velocity of  $B$  at time  $t = 0$ , giving your answer in terms of  $\mathbf{i}$  and  $\mathbf{j}$ . (2)

At time  $T_1$  hours after noon, both boats are moving with the same speed.

- (d) Find the exact value of  $T_1$  (4)

At time  $T_2$  hours after noon, both boats are moving in the same direction.

- (e) Show that  $3T_2^2 + pT_2 + q = 0$ , where  $p$  and  $q$  are integers to be found. (3)

16 marks

WME01/01 JANUARY 2026

## Question 6

Constant Acceleration in 2D

Also in Constant Acceleration in 2D

Primary: Working with Vectors

6. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin  $O$ .]

At 9 am, a ship  $S$  leaves a harbour and moves with constant velocity  $(10\mathbf{i} + 15\mathbf{j})\text{ km h}^{-1}$

- (a) Find the exact speed of  $S$

(2)

The harbour is at the origin  $O$ .

At 9 am, a boat  $B$  leaves the point with position vector  $(50\mathbf{i} + 10\mathbf{j})\text{ km}$  and moves with constant velocity  $(-8\mathbf{i} + 10\mathbf{j})\text{ km h}^{-1}$

- (b) Find the direction in which  $B$  is moving, giving your answer as a bearing to the nearest degree.

(3)

- (c) Show that at time  $t$  hours after 9 am,

$$\overrightarrow{SB} = [(50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}] \text{ km}$$

(4)

- (d) Show that  $B$  and  $S$  do not collide.

(3)

When  $B$  is north-east of  $S$ , the distance between  $S$  and  $B$  is  $d\text{ km}$ .

- (e) Find the value of  $d$ .

(4)

TOPIC

# Forces

## Question 6

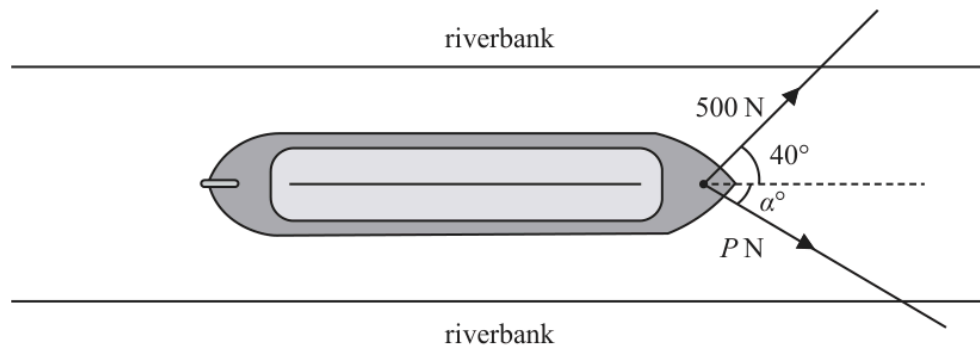


Figure 3

A boat is pulled along a river at a constant speed by two ropes.

The banks of the river are parallel and the boat travels horizontally in a straight line, parallel to the riverbanks.

- The tension in the first rope is 500 N acting at an angle of  $40^\circ$  to the direction of motion, as shown in Figure 3.
- The tension in the second rope is  $P$  newtons, acting at an angle of  $\alpha^\circ$  to the direction of motion, also shown in Figure 3.
- The resistance to motion of the boat as it moves through the water is a constant force of magnitude 900 N

The boat is modelled as a particle. The ropes are modelled as being light and lying in a horizontal plane.

Use the model to find

- the value of  $\alpha$
- the value of  $P$

(8)

## Question 6

6.

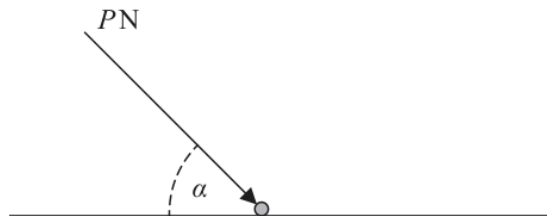


Figure 3

A particle of weight  $W$  newtons lies at rest on a rough horizontal surface, as shown in Figure 3.

A force of magnitude  $P$  newtons is applied to the particle.

The force acts at an angle  $\alpha$  to the horizontal, where  $\tan \alpha = \frac{4}{3}$

The coefficient of friction between the particle and the surface is  $\frac{1}{4}$

Given that the particle does not move, show that

$$P \leq \frac{5W}{8}$$

(7)



WME01/01 OCTOBER 2019

## Question 7

7 marks

Forces

Also in Forces

Primary: Working with Vectors

7. Two forces,  $\mathbf{F}$  and  $\mathbf{G}$ , act on a particle. The force  $\mathbf{F}$  has magnitude 4N and acts in a direction with a bearing of  $120^\circ$  and the force  $\mathbf{G}$  has magnitude 6N and acts due north.

Given that  $\mathbf{P} = 2\mathbf{F} + \mathbf{G}$ , find

- (i) the magnitude of  $\mathbf{P}$
- (ii) the direction of  $\mathbf{P}$ , giving your answer as a bearing to the nearest degree.

(7)

WME01/01 JANUARY 2020

## Question 7

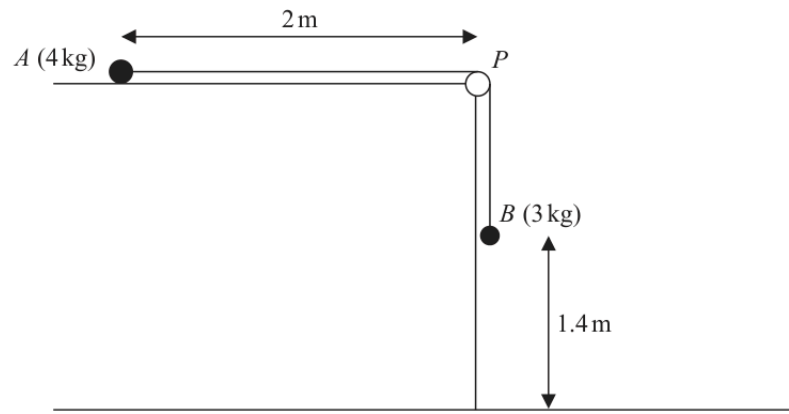
18 marks

Forces

Also in Forces

Primary: Newton's Second Law

7.



**Figure 3**

A particle  $A$  of mass  $4\text{ kg}$  is held at rest on a rough horizontal table. Particle  $A$  is attached to one end of a string that passes over a pulley  $P$ . The pulley is fixed at the edge of the table. The other end of the string is attached to a particle  $B$ , of mass  $3\text{ kg}$ , which hangs freely below  $P$ .

The part of the string from  $A$  to  $P$  is perpendicular to the edge of the table and  $A$ ,  $P$  and  $B$  all lie in the same vertical plane.

The string is modelled as being light and inextensible and the pulley is modelled as being small, smooth and light.

The system is released from rest with the string taut. At the instant of release,  $A$  is  $2\text{ m}$  from the edge of the table and  $B$  is  $1.4\text{ m}$  above a horizontal floor, as shown in Figure 3.

After descending with constant acceleration for  $2\text{ seconds}$ ,  $B$  hits the floor and does not rebound.

(a) Show that the acceleration of  $A$  before  $B$  hits the floor is  $0.7\text{ ms}^{-2}$  (2)

(b) State which of the modelling assumptions you have used in order to answer part (a). (1)

(c) Find the magnitude of the resultant force exerted on the pulley by the string. (4)

The coefficient of friction between  $A$  and the table is  $\mu$ .

(d) Find the value of  $\mu$ . (6)

(e) Determine, by calculation, whether or not  $A$  reaches the pulley. (5)

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**(Total 18 marks)**

## Question 6

Also in Forces

Primary: Resolving Forces, Inclined Planes

6.

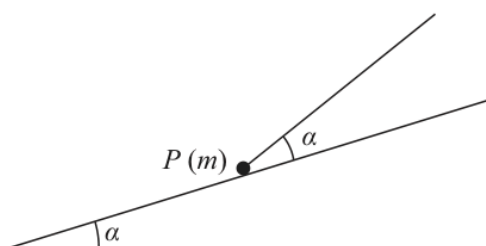


Figure 3

A particle  $P$  of mass  $m$  is held in equilibrium on a fixed rough inclined plane by a light inextensible string.

The plane is inclined at an angle  $\alpha$  to the horizontal, where  $\alpha < 45^\circ$

The string is inclined to the plane at angle  $\alpha$ , as shown in Figure 3.

The string lies in a vertical plane that contains a line of greatest slope of the inclined plane.

When the tension in the string is  $0.75mg$ ,  $P$  is on the point of moving up the plane.

- (a) Find an expression for the magnitude of the frictional force acting on  $P$ , giving your answer in terms of  $m$ ,  $g$  and  $\alpha$ .

(3)

The coefficient of friction between  $P$  and the plane is  $\frac{1}{2}$

- (b) Show that

$$\tan \alpha = \frac{2}{5}$$

(6)

The string breaks.

- (c) Determine whether  $P$  remains at rest. You must justify your reasoning.

(3)

TOPIC

# Newton's Second Law

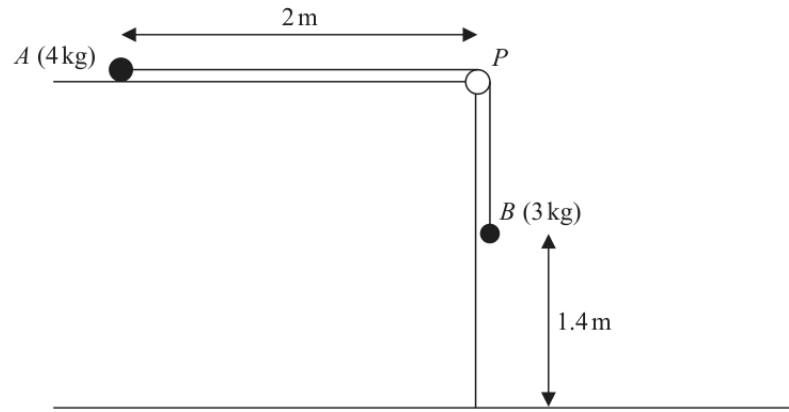
WME01/01 JANUARY 2020

18 marks

## Question 7

Newton's Second Law

7.



**Figure 3**

A particle  $A$  of mass  $4\text{ kg}$  is held at rest on a rough horizontal table. Particle  $A$  is attached to one end of a string that passes over a pulley  $P$ . The pulley is fixed at the edge of the table. The other end of the string is attached to a particle  $B$ , of mass  $3\text{ kg}$ , which hangs freely below  $P$ .

The part of the string from  $A$  to  $P$  is perpendicular to the edge of the table and  $A$ ,  $P$  and  $B$  all lie in the same vertical plane.

The string is modelled as being light and inextensible and the pulley is modelled as being small, smooth and light.

The system is released from rest with the string taut. At the instant of release,  $A$  is  $2\text{ m}$  from the edge of the table and  $B$  is  $1.4\text{ m}$  above a horizontal floor, as shown in Figure 3.

After descending with constant acceleration for  $2\text{ seconds}$ ,  $B$  hits the floor and does not rebound.

(a) Show that the acceleration of  $A$  before  $B$  hits the floor is  $0.7\text{ ms}^{-2}$  (2)

(b) State which of the modelling assumptions you have used in order to answer part (a). (1)

(c) Find the magnitude of the resultant force exerted on the pulley by the string. (4)

The coefficient of friction between  $A$  and the table is  $\mu$ .

(d) Find the value of  $\mu$ . (6)

(e) Determine, by calculation, whether or not  $A$  reaches the pulley. (5)

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**(Total 18 marks)**

## Question 7

## Newton's Second Law

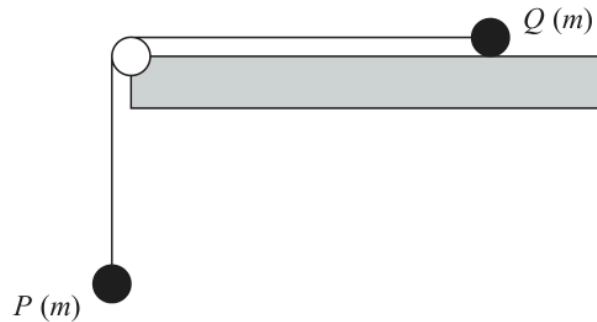


Figure 4

A particle  $P$  of mass  $m$  is attached to one end of a light inextensible string. Another particle  $Q$ , also of mass  $m$ , is attached to the other end of the string. The string passes over a small smooth pulley which is fixed at the edge of a rough horizontal table. Particle  $Q$  is held at rest on the table and particle  $P$  hangs vertically below the pulley with the string taut, as shown in Figure 4.

The pulley,  $P$  and  $Q$  all lie in the same vertical plane.

The coefficient of friction between  $Q$  and the table is  $\mu$ , where  $\mu < 1$

Particle  $Q$  is released from rest.

The tension in the string before  $Q$  hits the pulley is  $kmg$ , where  $k$  is a constant.

(a) Find  $k$  in terms of  $\mu$ .

(7)

Given that  $Q$  is initially a distance  $d$  from the pulley,

(b) find, in terms of  $d$ ,  $g$  and  $\mu$ , the time taken by  $Q$ , after release, to reach the pulley.

(4)

(c) Describe what would happen if  $\mu \geq 1$ , giving a reason for your answer.

(2)



## Question 7

## Newton's Second Law

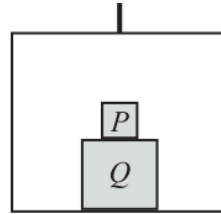


Figure 4

A simple lift operates by means of a vertical cable which is attached to the top of the lift.

The lift has mass  $m$

A box  $Q$  is placed on the floor of the lift.

A box  $P$  is placed directly on top of box  $Q$ , as shown in Figure 4.

The cable is modelled as being light and inextensible and air resistance is modelled as being negligible.

The tension in the cable is  $\frac{42mg}{5}$

The lift and its contents move vertically upwards with acceleration  $\frac{2g}{5}$

Using the model,

- (a) find, in terms of  $m$ , the combined mass of boxes  $P$  and  $Q$  (4)

During the motion of the lift, the force exerted on box  $P$  by box  $Q$  is  $\frac{14mg}{5}$

Using the model,

- (b) find, in terms of  $m$ , the mass of box  $P$  (3)

## Question 7

## Newton's Second Law

7.

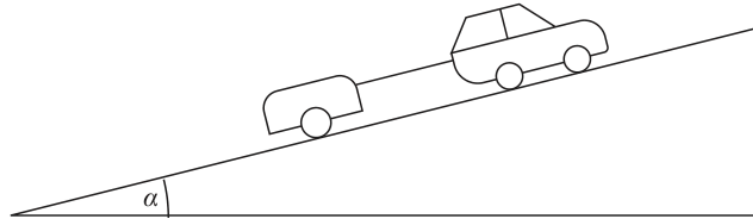


Figure 4

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle  $\alpha$  to the horizontal, where  $\sin \alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration  $0.75 \text{ m s}^{-2}$

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude  $2R$  newtons
- the **trailer** is modelled as a constant force of magnitude  $R$  newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of  $R$  is 60

(4)

(b) find the tension in the tow bar.

(3)

When the car and trailer are moving at a speed of  $12 \text{ m s}^{-1}$ , the tow bar breaks.

Given that the non-gravitational resistance to motion of the trailer remains unchanged,

(c) use the model to find the further distance moved by the trailer before it first comes to rest.

(4)

## Question 7

## Newton's Second Law

7.

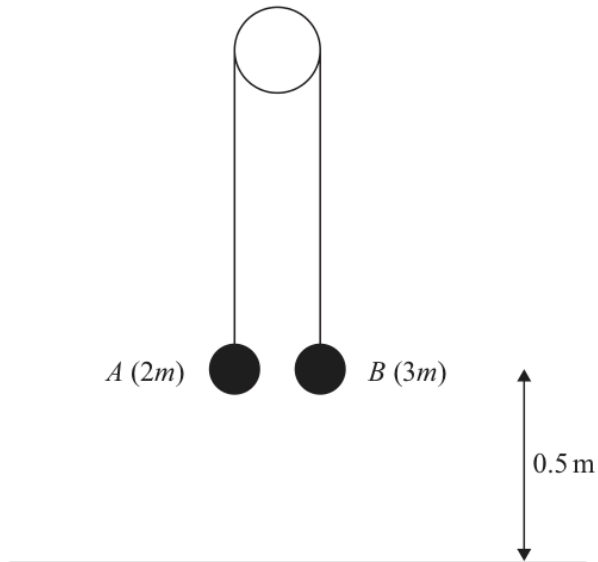


Figure 4

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end is attached to a particle  $B$  of mass  $3m$ . The string passes over a small smooth fixed pulley. The string is taut and both straight parts of the string are vertical. Both particles are held at a distance  $0.5\text{ m}$  above a horizontal surface, as shown in Figure 4.

The system is released from rest and  $B$  moves downwards.

- (a) Find the tension in the string in terms of  $m$  and  $g$ .

(5)

- (b) Find the speed of  $B$  at the instant it strikes the surface.

(4)

In the subsequent motion,  $A$  does not reach the pulley and  $B$  does not rebound after it strikes the surface.

- (c) Find the time from the instant when the system is released from rest to the instant when  $A$  first reaches a height of  $1.06\text{ m}$  above the surface.

(6)

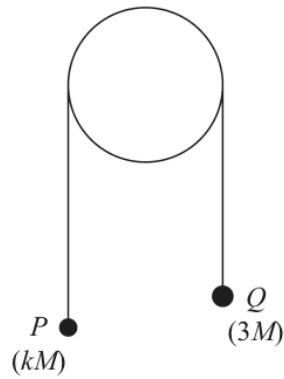
WME01/01 MAY/JUNE 2025

17 marks

## Question 8

Newton's Second Law

8.



**Figure 5**

Two small balls,  $P$  and  $Q$ , have masses  $kM$  and  $3M$  respectively, where  $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion,  $P$  moves with an acceleration of magnitude  $\frac{1}{5}g$

The balls are modelled as particles.

(a) Write down an equation of motion for  $P$ . (2)

(b) Find the value of  $k$ . (3)

Given that  $M = 0.5 \text{ kg}$ ,

(c) find the magnitude of the force exerted on the pulley by the string while  $Q$  is moving downwards. (3)

At the instant when the system is released,  $P$  is more than  $2.5 \text{ m}$  from the pulley and  $Q$  is  $2.5 \text{ m}$  above horizontal ground.

After hitting the ground,  $Q$  rebounds with a speed of  $0.4 \text{ m s}^{-1}$

(d) Find the magnitude of the impulse received by  $Q$  when it hits the ground. (5)

In the subsequent motion,  $P$  does not hit the pulley.

(e) Find the total time from when the balls are released until  $P$  first comes to rest. (4)

12 marks

WME01/01 OCTOBER 2019

## Question 6

Newton's Second Law

Also in Newton's Second Law

Primary: Kinematics Graphs

6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time  $t = 0$  seconds and moves with uniform acceleration  $0.8 \text{ m s}^{-2}$  for  $T$  seconds, reaching a speed of  $V \text{ m s}^{-1}$ . She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (a) Sketch a speed-time graph for the model of the motion of the athlete from the instant when she starts to the instant when she crosses the finishing line. (2)
- (b) Write down an expression for  $V$  in terms of  $T$ . (1)
- (c) Show that  $T^2 - kT + 500 = 0$ , where  $k$  is a constant to be found. (4)
- (d) Hence find the value of  $T$ , justifying your answer carefully. (3)
- (e) Considering your speed-time graph or otherwise, state two ways, apart from including air resistance, in which the model could be made to be more realistic. (2)

17 marks

WME01/01 JANUARY 2021

Newton's Second Law

## Question 8

Also in Newton's Second Law

Primary: Resolving Forces, Inclined Planes

8.

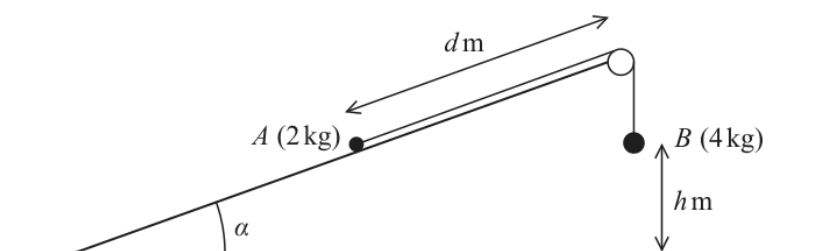


Figure 3

Two particles,  $A$  and  $B$ , have masses  $2\text{ kg}$  and  $4\text{ kg}$  respectively. The particles are connected by a light inextensible string. The string passes over a small smooth pulley which is fixed at the top of a rough plane. The plane is inclined to the horizontal ground at an angle  $\alpha$  where  $\tan \alpha = \frac{3}{4}$ . The particle  $A$  is held at rest on the plane at a distance  $d$  metres from the pulley. The particle  $B$  hangs freely at rest, vertically below the pulley, at a distance  $h$  metres above the ground, as shown in Figure 3. The part of the string between  $A$  and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between  $A$  and the plane is  $\frac{1}{4}$ .

The system is released from rest with the string taut and  $B$  descends.

- (a) Find the tension in the string as  $B$  descends.

(9)

On hitting the ground,  $B$  immediately comes to rest.

Given that  $A$  comes to rest before reaching the pulley,

- (b) find, in terms of  $h$ , the range of possible values of  $d$ .

(7)

- (c) State one physical factor, other than air resistance, that could be taken into account to make the model described above more realistic.

(1)

(Total 17 marks)

14 marks

WME01/01 OCTOBER 2021

Newton's Second Law

## Question 7

Also in Newton's Second Law

Primary: Resolving Forces, Inclined Planes

7.

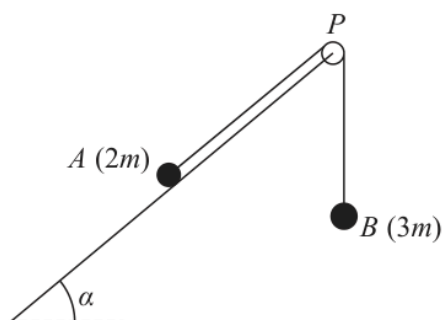


Figure 4

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . The string passes over a small, smooth, light pulley  $P$  which is fixed at the top of a rough inclined plane. The plane is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{3}{4}$

Particle  $A$  is held at rest on the plane with the string taut and  $B$  hanging freely below  $P$ , as shown in Figure 4. The section of the string  $AP$  is parallel to a line of greatest slope of the plane.

The coefficient of friction between  $A$  and the plane is  $\frac{1}{2}$

Particle  $A$  is released and begins to move up the plane.

For the motion before  $A$  reaches the pulley,

(a) (i) write down an equation of motion for  $A$ ,

(ii) write down an equation of motion for  $B$ ,

(4)

(b) find, in terms of  $g$ , the acceleration of  $A$ ,

(5)

(c) find the magnitude of the force exerted on the pulley by the string.

(4)

(d) State how you have used the information that  $P$  is a smooth pulley.

(1)



12 marks

WME01/01 JANUARY 2022

Newton's Second Law

## Question 6

Also in Newton's Second Law

Primary: Working with Vectors

6. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors.]

A particle  $P$  of mass  $2\text{ kg}$  moves under the action of two forces,  $(p\mathbf{i} + q\mathbf{j})\text{ N}$  and  $(2q\mathbf{i} + p\mathbf{j})\text{ N}$ , where  $p$  and  $q$  are constants.

Given that the acceleration of  $P$  is  $(\mathbf{i} - \mathbf{j})\text{ ms}^{-2}$

(a) find the value of  $p$  and the value of  $q$ . (5)

(b) Find the size of the angle between the direction of the acceleration and the vector  $\mathbf{j}$ . (2)

At time  $t = 0$ , the velocity of  $P$  is  $(3\mathbf{i} - 4\mathbf{j})\text{ ms}^{-1}$

At  $t = T$  seconds,  $P$  is moving in the direction of the vector  $(11\mathbf{i} - 13\mathbf{j})$ .

(c) Find the value of  $T$ . (5)

**13 marks**

**WME01/01 JANUARY  
2022**

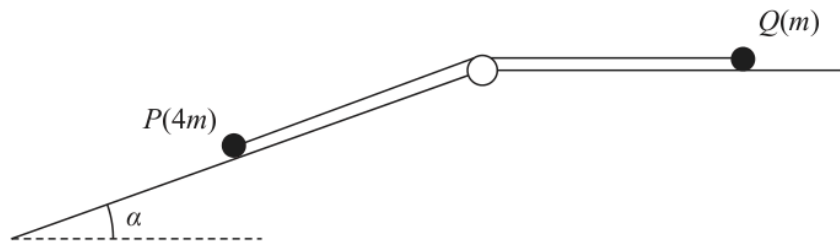
**Newton's Second Law**

## **Question 7**

**Also in Newton's Second Law**

**Primary: Resolving Forces, Inclined Planes**

7.



**Figure 4**

A particle  $P$  of mass  $4m$  lies on the surface of a fixed rough inclined plane.

The plane is inclined to the horizontal at an angle  $\alpha$  where  $\tan \alpha = \frac{3}{4}$

The particle  $P$  is attached to one end of a light inextensible string.

The string passes over a small smooth pulley that is fixed at the top of the plane. The other end of the string is attached to a particle  $Q$  of mass  $m$  which lies on a smooth horizontal plane.

The string lies along the horizontal plane and in the vertical plane that contains the pulley and a line of greatest slope of the inclined plane.

The system is released from rest with the string taut, as shown in Figure 4, and  $P$  moves down the plane.

The coefficient of friction between  $P$  and the plane is  $\frac{1}{4}$

For the motion before  $Q$  reaches the pulley

(a) write down an equation of motion for  $Q$ , (1)

(b) find, in terms of  $m$  and  $g$ , the tension in the string, (7)

(c) find the magnitude of the force exerted on the pulley by the string. (4)

(d) State where in your working you have used the information that the string is light. (1)

9 marks

WME01/01 OCTOBER 2022

Newton's Second Law

## Question 6

Also in Newton's Second Law

Primary: Working with Vectors

6. [In this question,  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors.]

A particle  $A$  of mass  $0.5\text{ kg}$  is at rest on a smooth horizontal plane.

At time  $t = 0$ , two forces,  $\mathbf{F}_1 = (-3\mathbf{i} + 2\mathbf{j})\text{ N}$  and  $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})\text{ N}$ , where  $p$  and  $q$  are constants, are applied to  $A$ .

Given that  $A$  moves in the direction of the vector  $(\mathbf{i} - 2\mathbf{j})$ ,

(a) show that  $2p + q - 4 = 0$  (4)

Given that  $p = 5$

(b) find the speed of  $A$  at time  $t = 4$  seconds. (5)

15 marks

WME01/01 JANUARY  
2023

Newton's Second Law

## Question 8

Also in Newton's Second Law

Primary: Resolving Forces, Inclined Planes

8.

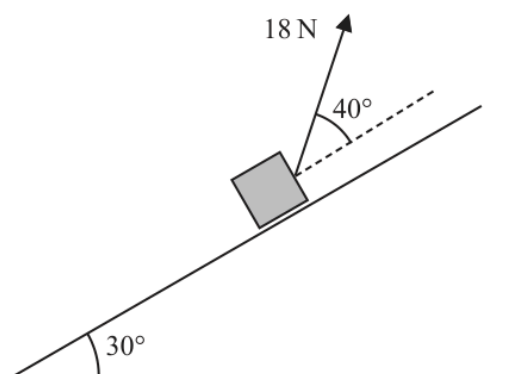


Figure 5

A parcel of mass  $2\text{ kg}$  is pulled up a rough inclined plane by the action of a constant force.

The force has magnitude  $18\text{ N}$  and acts at an angle of  $40^\circ$  to the plane.

The line of action of the force lies in a vertical plane containing a line of greatest slope of the inclined plane.

The plane is inclined at an angle of  $30^\circ$  to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is  $0.3$

The parcel is modelled as a particle  $P$

(a) Find the acceleration of  $P$  (8)

The points  $A$  and  $B$  lie on a line of greatest slope of the plane, where  $AB = 5\text{ m}$  and  $B$  is above  $A$ . Particle  $P$  passes through  $A$  with speed  $2\text{ m s}^{-1}$  in the direction  $AB$ .

(b) Find the speed of  $P$  as it passes through  $B$ . (3)

The force of  $18\text{ N}$  is removed at the instant  $P$  passes through  $B$ . As a result,  $P$  comes to rest at the point  $C$ .

(c) Determine whether  $P$  will remain at rest at  $C$ . You must show all stages of your working clearly. (4)

7 marks

WME01/01 MAY/JUNE 2023

Newton's Second Law

## Question 6

Also in Newton's Second Law

Primary: Forces

6.

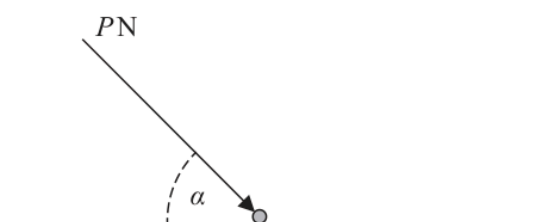


Figure 3

A particle of weight  $W$  newtons lies at rest on a rough horizontal surface, as shown in Figure 3.

A force of magnitude  $P$  newtons is applied to the particle.

The force acts at an angle  $\alpha$  to the horizontal, where  $\tan \alpha = \frac{4}{3}$

The coefficient of friction between the particle and the surface is  $\frac{1}{4}$

Given that the particle does not move, show that

$$P \leq \frac{5W}{8}$$

(7)

13 marks

WME01/01 OCTOBER 2023

Newton's Second Law

## Question 7

Also in Newton's Second Law

Primary: Resolving Forces, Inclined Planes

7.

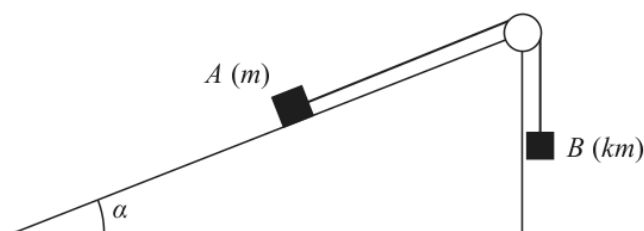


Figure 4

Figure 4 shows a block  $A$  of mass  $m$  held at rest on a rough plane. The plane is inclined at an angle  $\alpha$  to the horizontal and the coefficient of friction between the block and the plane is  $\mu$ .

One end of a light inextensible string is now attached to  $A$ . The string passes over a small smooth pulley which is fixed at the top of the plane. The other end of the string is attached to a block  $B$  of mass  $km$ . Block  $B$  hangs vertically below the pulley, with the string taut.

The string from  $A$  to the pulley lies along a line of greatest slope of the plane.

Both  $A$  and  $B$  are modelled as particles.

When the system is released from rest,  $A$  moves up the plane and the tension in the string is  $\frac{4mg}{3}$

Given that  $\mu = \frac{1}{3}$  and  $\tan \alpha = \frac{7}{24}$

- (a) (i) find the magnitude of the acceleration of  $A$ , giving your answer in terms of  $g$ ,  
 (ii) find the value of  $k$ .

(9)

- (b) Find the magnitude of the resultant force exerted on the pulley by the string, giving your answer in terms of  $m$  and  $g$ .

(4)

15 marks

WME01/01 MAY/JUNE  
2024

Newton's Second Law

## Question 8

Also in Newton's Second Law

Primary: Resolving Forces, Inclined Planes

8.

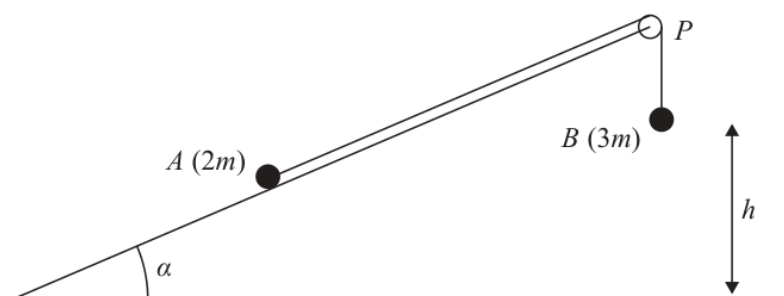


Figure 4

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . Particle  $A$  is held at rest on a rough plane which is inclined to horizontal ground at an angle  $\alpha$ , where  $\tan \alpha = \frac{5}{12}$ .

The string passes over a small smooth pulley  $P$  which is fixed at the top of the plane. Particle  $B$  hangs vertically below  $P$  with the string taut, at a height  $h$  above the ground, as shown in Figure 4.

The part of the string between  $A$  and  $P$  lies along a line of greatest slope of the plane. The two particles, the string and the pulley all lie in the same vertical plane.

The coefficient of friction between  $A$  and the plane is  $\frac{11}{36}$ .

The particle  $A$  is released from rest and begins to move up the plane.

(a) Show that the frictional force acting on  $A$  as it moves up the plane is  $\frac{22mg}{39}$  (3)

(b) Write down an equation of motion for  $B$ . (2)

(c) Show that the acceleration of  $A$  immediately after its release is  $\frac{1}{3}g$  (4)

In the subsequent motion,  $A$  comes to rest before it reaches the pulley.

(d) Find, in terms of  $h$ , the total distance travelled by  $A$  from when it was released from rest to when it first comes to rest again. (6)



14 marks

WME01/01 OCTOBER 2025

Newton's Second Law

## Question 7

Also in Newton's Second Law

Primary: Resolving Forces, Inclined Planes

7.

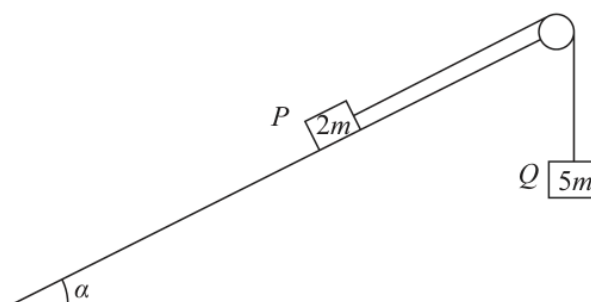


Figure 5

A block  $P$  of mass  $2m$  is held at rest on a fixed rough plane.

The plane is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{5}{12}$

One end of a light inextensible string is attached to  $P$ .

The string is parallel to a line of greatest slope of the plane and passes over a smooth light pulley which is fixed at the top of the plane.

The other end of the string is attached to a block  $Q$  of mass  $5m$ .

Block  $Q$  hangs vertically below the pulley, as shown in Figure 5.

The system is released from rest with the string taut and block  $P$  moves up the plane.

Immediately after the system is released, the tension in the string is  $T$  and the acceleration of the blocks is  $a$ .

The blocks are modelled as particles and air resistance is ignored.

(a) Write down an equation of motion for  $Q$ .

(2)

The coefficient of friction between  $P$  and the plane is  $\frac{1}{8}$

(b) Find  $T$  in terms of  $m$  and  $g$ .

(7)

(c) State how the solution to part (b) uses the fact that the string is inextensible.

(1)

The magnitude of the force exerted on the pulley by the string is  $kmg$ .

(d) Find the value of  $k$  to 3 significant figures.

(4)

**14 marks**

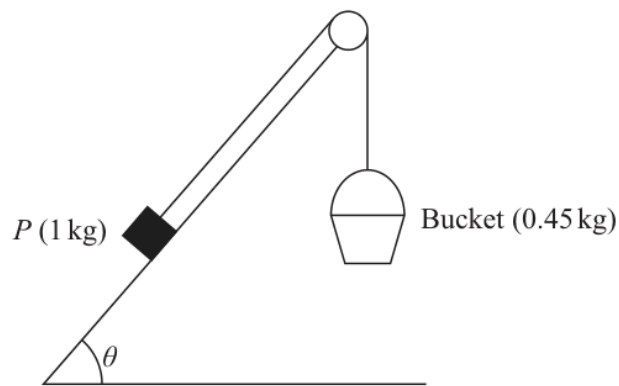
**WME01/01 JANUARY  
2026**

**Newton's Second Law**

## **Question 7**

**Also in Newton's Second Law**

**Primary: Resolving Forces, Inclined Planes**



**Figure 3**

A rough plane is inclined at an angle  $\theta$  to the horizontal, where  $\tan \theta = \frac{4}{3}$

One end of a light inextensible string is attached to a package  $P$ .

The string passes over a smooth pulley that is fixed at the top of the plane.

The string from  $P$  to the pulley lies along a line of greatest slope of the plane.

The other end of the string is attached to a bucket.

The package  $P$  has mass 1 kg and is **held** at rest on the plane.

The bucket has mass 0.45 kg and hangs vertically below the pulley with the string taut, as shown in Figure 3.

The coefficient of friction between  $P$  and the plane is  $\mu$

When a small block of mass 0.2 kg is placed in the bucket, the system is released and remains at rest in **limiting equilibrium**, with  $P$  on the point of slipping down the plane.

- (a) Show that  $\mu = \frac{1}{4}$  (7)

Additional identical blocks of mass 0.2 kg are added one at a time into the bucket until  $P$  starts to move **up** the plane.

At the instant when  $P$  starts to move up the plane, there are a **total** of  $n$  blocks in the bucket.

- (b) Find
- (i) the value of  $n$
  - (ii) the magnitude of the initial acceleration of  $P$
- (7)

**TOPIC**

# **Resolving Forces, Inclined Planes**

## Question 8

## Resolving Forces, Inclined Planes

8.

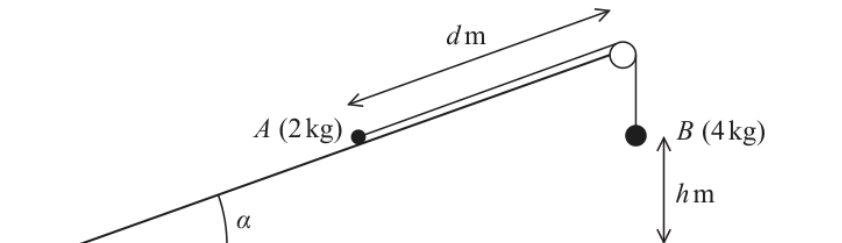


Figure 3

Two particles,  $A$  and  $B$ , have masses  $2\text{ kg}$  and  $4\text{ kg}$  respectively. The particles are connected by a light inextensible string. The string passes over a small smooth pulley which is fixed at the top of a rough plane. The plane is inclined to the horizontal ground at an angle  $\alpha$  where  $\tan \alpha = \frac{3}{4}$ . The particle  $A$  is held at rest on the plane at a distance  $d$  metres from the pulley. The particle  $B$  hangs freely at rest, vertically below the pulley, at a distance  $h$  metres above the ground, as shown in Figure 3. The part of the string between  $A$  and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between  $A$  and the plane is  $\frac{1}{4}$ .

The system is released from rest with the string taut and  $B$  descends.

- (a) Find the tension in the string as  $B$  descends.

(9)

On hitting the ground,  $B$  immediately comes to rest.

Given that  $A$  comes to rest before reaching the pulley,

- (b) find, in terms of  $h$ , the range of possible values of  $d$ .

(7)

- (c) State one physical factor, other than air resistance, that could be taken into account to make the model described above more realistic.

(1)

(Total 17 marks)

## Question 6

## Resolving Forces, Inclined Planes

6. A fixed rough plane is inclined at an angle  $\theta$  to the horizontal, where  $\tan \theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed  $5 \text{ m s}^{-1}$  from a point  $A$  on the plane, up a line of greatest slope of the plane.

The coefficient of friction between the particle and the plane is  $\frac{1}{4}$

- (a) Find the magnitude of the frictional force acting on the particle as it moves up the plane. (3)

The particle comes to instantaneous rest at the point  $B$ .

- (b) Find the distance  $AB$ . (5)

The particle now slides down the plane from  $B$ . At the instant when the particle passes through the point  $C$  on the plane, the speed of the particle is again  $5 \text{ m s}^{-1}$

- (c) Find the distance  $BC$ . (5)

## Question 7

## Resolving Forces, Inclined Planes

7.

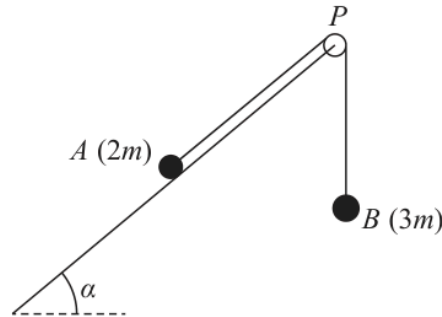


Figure 4

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . The string passes over a small, smooth, light pulley  $P$  which is fixed at the top of a rough inclined plane. The plane is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{3}{4}$

Particle  $A$  is held at rest on the plane with the string taut and  $B$  hanging freely below  $P$ , as shown in Figure 4. The section of the string  $AP$  is parallel to a line of greatest slope of the plane.

The coefficient of friction between  $A$  and the plane is  $\frac{1}{2}$

Particle  $A$  is released and begins to move up the plane.

For the motion before  $A$  reaches the pulley,

(a) (i) write down an equation of motion for  $A$ ,

(ii) write down an equation of motion for  $B$ ,

(4)

(b) find, in terms of  $g$ , the acceleration of  $A$ ,

(5)

(c) find the magnitude of the force exerted on the pulley by the string.

(4)

(d) State how you have used the information that  $P$  is a smooth pulley.

(1)

## Question 7

## Resolving Forces, Inclined Planes

7.

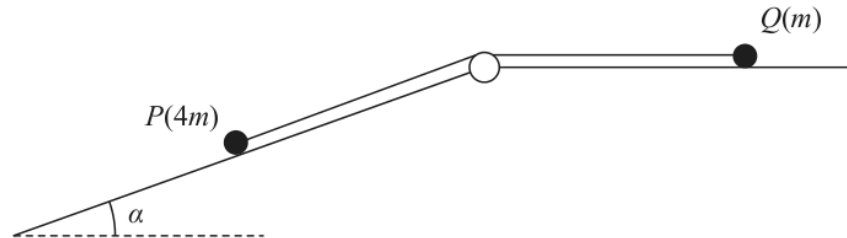


Figure 4

A particle  $P$  of mass  $4m$  lies on the surface of a fixed rough inclined plane.

The plane is inclined to the horizontal at an angle  $\alpha$  where  $\tan \alpha = \frac{3}{4}$

The particle  $P$  is attached to one end of a light inextensible string.

The string passes over a small smooth pulley that is fixed at the top of the plane. The other end of the string is attached to a particle  $Q$  of mass  $m$  which lies on a smooth horizontal plane.

The string lies along the horizontal plane and in the vertical plane that contains the pulley and a line of greatest slope of the inclined plane.

The system is released from rest with the string taut, as shown in Figure 4, and  $P$  moves down the plane.

The coefficient of friction between  $P$  and the plane is  $\frac{1}{4}$

For the motion before  $Q$  reaches the pulley

(a) write down an equation of motion for  $Q$ , (1)

(b) find, in terms of  $m$  and  $g$ , the tension in the string, (7)

(c) find the magnitude of the force exerted on the pulley by the string. (4)

(d) State where in your working you have used the information that the string is light. (1)



## Question 8

## Resolving Forces, Inclined Planes

8.

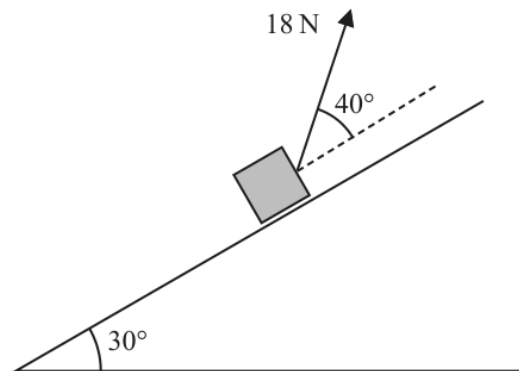


Figure 5

A parcel of mass  $2\text{ kg}$  is pulled up a rough inclined plane by the action of a constant force.

The force has magnitude  $18\text{ N}$  and acts at an angle of  $40^\circ$  to the plane.

The line of action of the force lies in a vertical plane containing a line of greatest slope of the inclined plane.

The plane is inclined at an angle of  $30^\circ$  to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is  $0.3$

The parcel is modelled as a particle  $P$

(a) Find the acceleration of  $P$

(8)

The points  $A$  and  $B$  lie on a line of greatest slope of the plane, where  $AB = 5\text{ m}$  and  $B$  is above  $A$ . Particle  $P$  passes through  $A$  with speed  $2\text{ m s}^{-1}$  in the direction  $AB$ .

(b) Find the speed of  $P$  as it passes through  $B$ .

(3)

The force of  $18\text{ N}$  is removed at the instant  $P$  passes through  $B$ . As a result,  $P$  comes to rest at the point  $C$ .

(c) Determine whether  $P$  will remain at rest at  $C$ . You must show all stages of your working clearly.

(4)

## Question 7

## Resolving Forces, Inclined Planes

7.

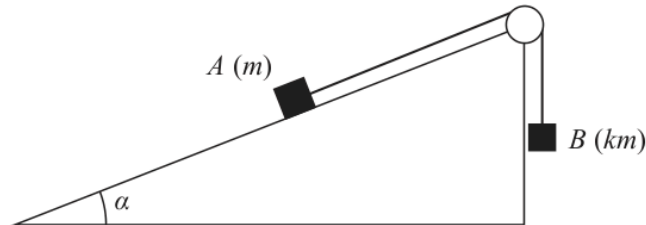


Figure 4

Figure 4 shows a block  $A$  of mass  $m$  held at rest on a rough plane. The plane is inclined at an angle  $\alpha$  to the horizontal and the coefficient of friction between the block and the plane is  $\mu$ .

One end of a light inextensible string is now attached to  $A$ . The string passes over a small smooth pulley which is fixed at the top of the plane. The other end of the string is attached to a block  $B$  of mass  $km$ . Block  $B$  hangs vertically below the pulley, with the string taut.

The string from  $A$  to the pulley lies along a line of greatest slope of the plane.

Both  $A$  and  $B$  are modelled as particles.

When the system is released from rest,  $A$  moves up the plane and the tension in the string is  $\frac{4mg}{3}$

Given that  $\mu = \frac{1}{3}$  and  $\tan \alpha = \frac{7}{24}$

- (a) (i) find the magnitude of the acceleration of  $A$ , giving your answer in terms of  $g$ ,  
 (ii) find the value of  $k$ .

(9)

- (b) Find the magnitude of the resultant force exerted on the pulley by the string, giving your answer in terms of  $m$  and  $g$ .

(4)

## Question 8

## Resolving Forces, Inclined Planes

8.

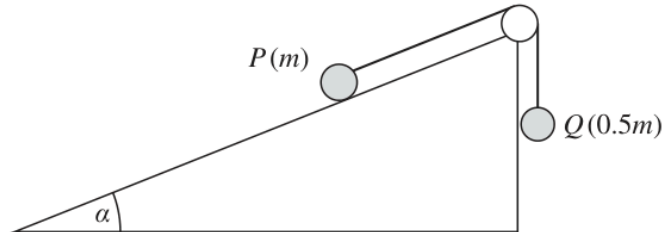


Figure 5

A fixed rough plane is inclined at an angle  $\alpha$  to the horizontal, where  $\tan \alpha = \frac{5}{12}$

A small smooth pulley is fixed at the top of the plane.

One end of a light inextensible string is attached to a particle  $P$  which is at rest on the plane. The string passes over the pulley and the other end of the string is attached to a particle  $Q$  which hangs vertically below the pulley, as shown in Figure 5.

Particle  $P$  has mass  $m$  and particle  $Q$  has mass  $0.5m$

The string from  $P$  to the pulley lies along a line of greatest slope of the plane.

The coefficient of friction between  $P$  and the plane is  $\mu$ .

The system is in **limiting equilibrium** with the string taut and  $P$  is on the point of slipping **up** the plane.

(a) Find the value of  $\mu$ .

(8)

The string breaks and  $P$  begins to move down the plane.

When particle  $P$  has travelled a distance of 0.8 m down the plane, the speed of  $P$  is  $V \text{ m s}^{-1}$

(b) Find the value of  $V$ .

(4)

## Question 6

## Resolving Forces, Inclined Planes

6.

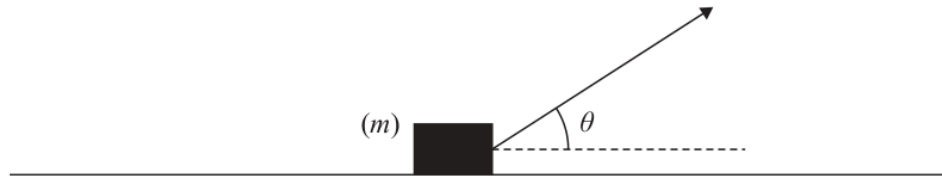


Figure 3

A box of mass  $m$  lies on a rough horizontal plane. The box is pulled along the plane in a straight line at **constant speed** by a light rope. The rope is inclined at an angle  $\theta$  to the plane, as shown in Figure 3.

The coefficient of friction between the box and the plane is  $\frac{1}{3}$

The box is modelled as a particle.

Given that  $\tan\theta = \frac{3}{4}$

(a) find, in terms of  $m$  and  $g$ , the tension in the rope.

(7)

The rope is now removed and the box is placed at rest on the plane.  
The box is then projected horizontally along the plane with speed  $u$ .

The box is again modelled as a particle.

When the box has moved a distance  $d$  along the plane, the speed of the box is  $\frac{1}{2}u$ .

(b) Find  $d$  in terms of  $u$  and  $g$ .

(5)

## Question 8

## Resolving Forces, Inclined Planes

8.

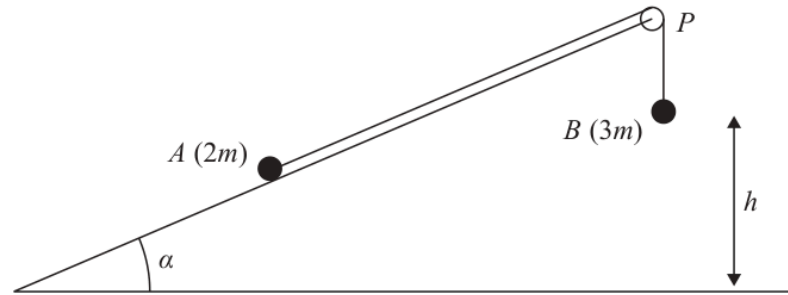


Figure 4

One end of a light inextensible string is attached to a particle  $A$  of mass  $2m$ . The other end of the string is attached to a particle  $B$  of mass  $3m$ . Particle  $A$  is held at rest on a rough plane which is inclined to horizontal ground at an angle  $\alpha$ , where  $\tan \alpha = \frac{5}{12}$ .

The string passes over a small smooth pulley  $P$  which is fixed at the top of the plane. Particle  $B$  hangs vertically below  $P$  with the string taut, at a height  $h$  above the ground, as shown in Figure 4.

The part of the string between  $A$  and  $P$  lies along a line of greatest slope of the plane. The two particles, the string and the pulley all lie in the same vertical plane.

The coefficient of friction between  $A$  and the plane is  $\frac{11}{36}$ .

The particle  $A$  is released from rest and begins to move up the plane.

(a) Show that the frictional force acting on  $A$  as it moves up the plane is  $\frac{22mg}{39}$  (3)

(b) Write down an equation of motion for  $B$ . (2)

(c) Show that the acceleration of  $A$  immediately after its release is  $\frac{1}{3}g$  (4)

In the subsequent motion,  $A$  comes to rest before it reaches the pulley.

(d) Find, in terms of  $h$ , the total distance travelled by  $A$  from when it was released from rest to when it first comes to rest again. (6)

## Question 6

## Resolving Forces, Inclined Planes

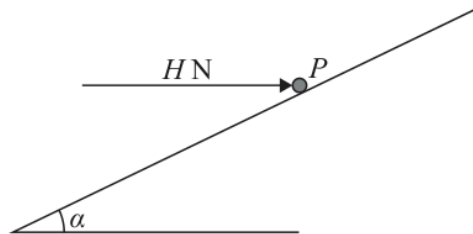


Figure 4

A particle  $P$  of mass  $5\text{ kg}$  lies on the surface of a rough plane.

The plane is inclined at an angle  $\alpha$  to the horizontal, where  $\tan \alpha = \frac{3}{4}$

The particle is held in equilibrium by a horizontal force of magnitude  $H$  newtons, as shown in Figure 4.

The horizontal force acts in a vertical plane containing a line of greatest slope of the inclined plane.

The coefficient of friction between the particle and the plane is  $\frac{1}{4}$

(a) Find the smallest possible value of  $H$ .

(6)

The horizontal force is now removed, and  $P$  starts to slide down the slope.

In the first  $T$  seconds after  $P$  is released from rest,  $P$  slides  $1.5\text{ m}$  down the slope.

(b) Find the value of  $T$ .

(6)

## Question 6

## Resolving Forces, Inclined Planes

6.

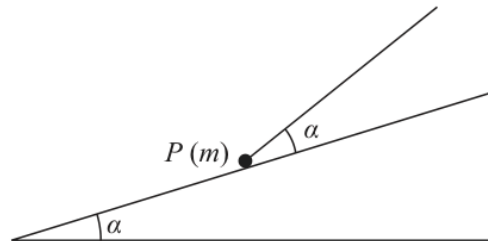


Figure 3

A particle  $P$  of mass  $m$  is held in equilibrium on a fixed rough inclined plane by a light inextensible string.

The plane is inclined at an angle  $\alpha$  to the horizontal, where  $\alpha < 45^\circ$

The string is inclined to the plane at angle  $\alpha$ , as shown in Figure 3.

The string lies in a vertical plane that contains a line of greatest slope of the inclined plane.

When the tension in the string is  $0.75mg$ ,  $P$  is on the point of moving up the plane.

- (a) Find an expression for the magnitude of the frictional force acting on  $P$ , giving your answer in terms of  $m$ ,  $g$  and  $\alpha$ .

(3)

The coefficient of friction between  $P$  and the plane is  $\frac{1}{2}$

- (b) Show that

$$\tan \alpha = \frac{2}{5}$$

(6)

The string breaks.

- (c) Determine whether  $P$  remains at rest. You must justify your reasoning.

(3)

## Question 6

## Resolving Forces, Inclined Planes

6.

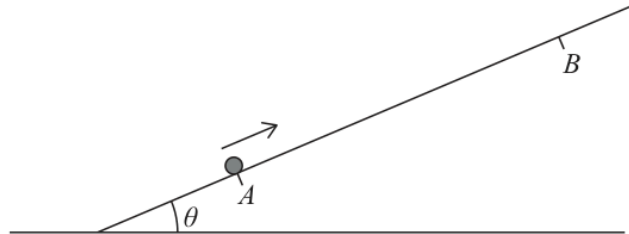


Figure 4

Figure 4 shows a rough plane inclined at an angle  $\theta$  to the horizontal, where  $\tan \theta = \frac{3}{4}$ .

The points  $A$  and  $B$  lie on a line of greatest slope of the plane, with  $B$  above  $A$ .

A package  $P$  of mass  $m$  is projected up the plane from  $A$  towards  $B$ .

The coefficient of friction between the plane and the package is  $\frac{1}{4}$ .

The package  $P$  is modelled as a particle.

- (a) Show that the **deceleration** of  $P$ , as it moves from  $A$  to  $B$ , is  $\frac{4}{5}g$  (6)

The package  $P$  comes to rest at  $B$ .

Given that  $P$  is projected from  $A$  with speed  $U \text{ m s}^{-1}$  and that  $AB = 1.5 \text{ m}$ ,

- (b) find the value of  $U$ . (2)

On reaching  $B$ ,  $P$  is held at rest there by a force of magnitude  $X$  newtons acting up the plane in the direction  $AB$ .

Given that the mass of  $P$  is  $2 \text{ kg}$ ,

- (c) find the smallest possible value of  $X$ . (3)



## Question 7

## Resolving Forces, Inclined Planes

7.

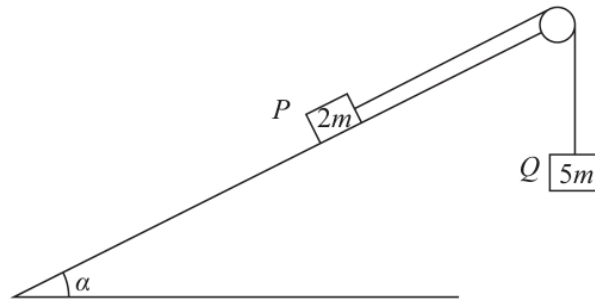


Figure 5

A block  $P$  of mass  $2m$  is held at rest on a fixed rough plane.

The plane is inclined to the horizontal at an angle  $\alpha$ , where  $\tan \alpha = \frac{5}{12}$

One end of a light inextensible string is attached to  $P$ .

The string is parallel to a line of greatest slope of the plane and passes over a smooth light pulley which is fixed at the top of the plane.

The other end of the string is attached to a block  $Q$  of mass  $5m$ .

Block  $Q$  hangs vertically below the pulley, as shown in Figure 5.

The system is released from rest with the string taut and block  $P$  moves up the plane.

Immediately after the system is released, the tension in the string is  $T$  and the acceleration of the blocks is  $a$ .

The blocks are modelled as particles and air resistance is ignored.

(a) Write down an equation of motion for  $Q$ .

(2)

The coefficient of friction between  $P$  and the plane is  $\frac{1}{8}$

(b) Find  $T$  in terms of  $m$  and  $g$ .

(7)

(c) State how the solution to part (b) uses the fact that the string is inextensible.

(1)

The magnitude of the force exerted on the pulley by the string is  $kmg$ .

(d) Find the value of  $k$  to 3 significant figures.

(4)

## Question 7

## Resolving Forces, Inclined Planes

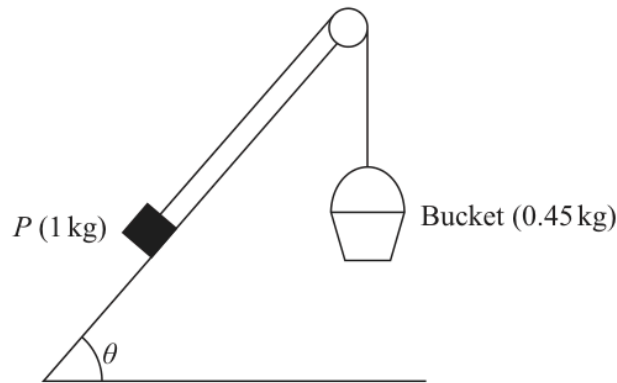


Figure 3

A rough plane is inclined at an angle  $\theta$  to the horizontal, where  $\tan \theta = \frac{4}{3}$

One end of a light inextensible string is attached to a package  $P$ .

The string passes over a smooth pulley that is fixed at the top of the plane.

The string from  $P$  to the pulley lies along a line of greatest slope of the plane.

The other end of the string is attached to a bucket.

The package  $P$  has mass 1 kg and is **held** at rest on the plane.

The bucket has mass 0.45 kg and hangs vertically below the pulley with the string taut, as shown in Figure 3.

The coefficient of friction between  $P$  and the plane is  $\mu$

When a small block of mass 0.2 kg is placed in the bucket, the system is released and remains at rest in **limiting equilibrium**, with  $P$  on the point of slipping down the plane.

- (a) Show that  $\mu = \frac{1}{4}$  (7)

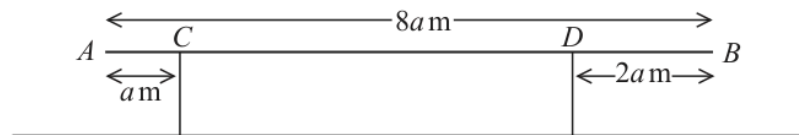
Additional identical blocks of mass 0.2 kg are added one at a time into the bucket until  $P$  starts to move **up** the plane.

At the instant when  $P$  starts to move up the plane, there are a **total** of  $n$  blocks in the bucket.

- (b) Find
- (i) the value of  $n$
  - (ii) the magnitude of the initial acceleration of  $P$
- (7)

**10 marks****WME01/01 MAY/JUNE 2021****Resolving Forces, Inclined Planes**

## Question 7

**Also in Resolving Forces, Inclined Planes****Primary: Moments****7.****Figure 2**

A non-uniform beam  $AB$ , of mass  $60$  kg and length  $8a$  metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at  $C$ , where  $AC = a$  metres and the other support is at  $D$ , where  $DB = 2a$  metres, as shown in Figure 2.

The magnitude of the normal reaction between the beam and the support at  $D$  is three times the magnitude of the normal reaction between the beam and the support at  $C$ .

By modelling the beam as a non-uniform rod whose centre of mass is at a distance  $x$  metres from  $A$ ,

- (a) find an expression for  $x$  in terms of  $a$ .

**(5)**

A box of mass  $M$  kg is placed on the beam at  $E$ , where  $AE = 2a$  metres.

The beam remains in equilibrium in a horizontal position.

The magnitude of the normal reaction between the beam and the support at  $C$  is now equal to the magnitude of the normal reaction between the beam and the support at  $D$ .

By modelling the box as a particle,

- (b) find the value of  $M$ .

**(5)**

13 marks

WME01/01 OCTOBER 2022

## Question 7

Resolving Forces, Inclined Planes

Also in Resolving Forces, Inclined Planes

Primary: Newton's Second Law

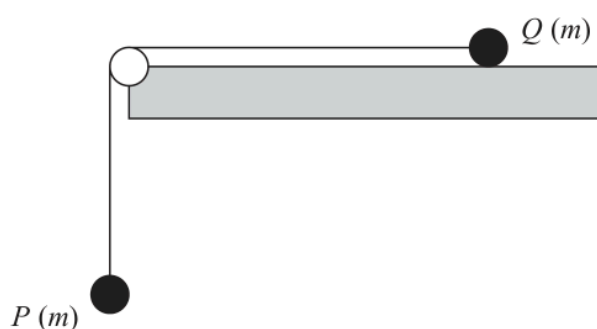


Figure 4

A particle  $P$  of mass  $m$  is attached to one end of a light inextensible string. Another particle  $Q$ , also of mass  $m$ , is attached to the other end of the string. The string passes over a small smooth pulley which is fixed at the edge of a rough horizontal table. Particle  $Q$  is held at rest on the table and particle  $P$  hangs vertically below the pulley with the string taut, as shown in Figure 4.

The pulley,  $P$  and  $Q$  all lie in the same vertical plane.

The coefficient of friction between  $Q$  and the table is  $\mu$ , where  $\mu < 1$

Particle  $Q$  is released from rest.

The tension in the string before  $Q$  hits the pulley is  $kmg$ , where  $k$  is a constant.

(a) Find  $k$  in terms of  $\mu$ .

(7)

Given that  $Q$  is initially a distance  $d$  from the pulley,

(b) find, in terms of  $d$ ,  $g$  and  $\mu$ , the time taken by  $Q$ , after release, to reach the pulley.

(4)

(c) Describe what would happen if  $\mu \geq 1$ , giving a reason for your answer.

(2)

11 marks

WME01/01 MAY/JUNE 2023

## Question 7

Resolving Forces, Inclined Planes

Also in Resolving Forces, Inclined Planes

Primary: Newton's Second Law

7.

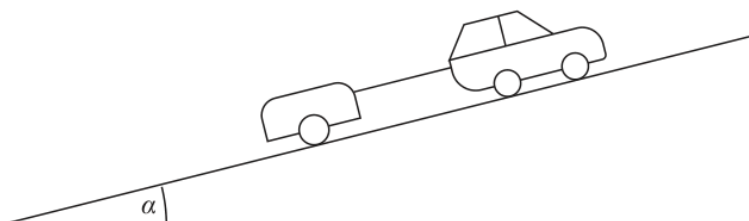


Figure 4

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle  $\alpha$  to the horizontal, where  $\sin \alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration  $0.75 \text{ m s}^{-2}$

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude  $2R$  newtons
- the **trailer** is modelled as a constant force of magnitude  $R$  newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of  $R$  is 60

(4)

(b) find the tension in the tow bar.

(3)

When the car and trailer are moving at a speed of  $12 \text{ m s}^{-1}$ , the tow bar breaks.

Given that the non-gravitational resistance to motion of the trailer remains unchanged,

(c) use the model to find the further distance moved by the trailer before it first comes to rest.

(4)

**TOPIC**

# **Momentum, Impulse & Collisions**

12 marks

WME01/01 JANUARY 2024

Momentum, Impulse & Collisions

## Question 6

Also in Momentum, Impulse & Collisions

Primary: Constant Acceleration in 1D

6. A particle is projected vertically upwards from a point  $A$  with speed  $24 \text{ m s}^{-1}$

The point  $A$  is  $2.5 \text{ m}$  vertically above the point  $B$ .

Point  $B$  lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at  $B$  with speed  $V \text{ m s}^{-1}$

After hitting the ground the particle does not rebound.

- (a) Find the value of  $V$ .

(3)

- (b) Find the time taken for the particle to reach  $B$ .

(3)

The point  $C$  is  $10 \text{ m}$  vertically above  $A$ .

- (c) Find the length of time for which the particle is above  $C$ .

(4)

- (d) Sketch a speed-time graph for the motion of the particle from projection to the instant that it reaches  $B$ . (No further calculations are required.)

(2)

11 marks

WME01/01 OCTOBER 2024

Momentum, Impulse & Collisions

## Question 7

Also in Momentum, Impulse & Collisions

Primary: Constant Acceleration in 1D

- 7 At time  $t = 0$ , a small ball  $A$  is projected vertically upwards with speed  $8 \text{ m s}^{-1}$  from a fixed point on horizontal ground.

The ball hits the ground again for the first time at time  $t = T_1$  seconds.

Ball  $A$  is modelled as a particle moving freely under gravity.

- (a) Show that  $T_1 = 1.63$  to 3 significant figures.

(2)

After the first impact with the ground,  $A$  rebounds to a height of 2 m above the ground.

Given that the mass of  $A$  is 0.1 kg,

- (b) find the magnitude of the impulse received by  $A$  as a result of its first impact with the ground.

(5)

At time  $t = 1$  second, another small ball  $B$  is projected vertically upwards from another point on the ground with speed  $5 \text{ m s}^{-1}$

Ball  $B$  is modelled as a particle moving freely under gravity.

At time  $t = T_2$  seconds ( $T_2 > 1$ ),  $A$  and  $B$  are at the same height above the ground for the first time.

- (c) Find the value of  $T_2$

(4)



**17 marks**

**WME01/01 MAY/JUNE 2025**

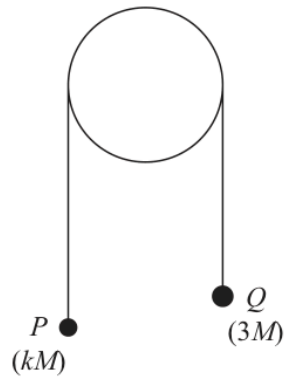
## **Question 8**

**Momentum, Impulse & Collisions**

**Also in Momentum, Impulse & Collisions**

**Primary: Newton's Second Law**

8.



**Figure 5**

Two small balls,  $P$  and  $Q$ , have masses  $kM$  and  $3M$  respectively, where  $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion,  $P$  moves with an acceleration of magnitude  $\frac{1}{5}g$

The balls are modelled as particles.

(a) Write down an equation of motion for  $P$ .

(2)

(b) Find the value of  $k$ .

(3)

Given that  $M = 0.5 \text{ kg}$ ,

(c) find the magnitude of the force exerted on the pulley by the string while  $Q$  is moving downwards.

(3)

At the instant when the system is released,  $P$  is more than  $2.5 \text{ m}$  from the pulley and  $Q$  is  $2.5 \text{ m}$  above horizontal ground.

After hitting the ground,  $Q$  rebounds with a speed of  $0.4 \text{ m s}^{-1}$

(d) Find the magnitude of the impulse received by  $Q$  when it hits the ground.

(5)

In the subsequent motion,  $P$  does not hit the pulley.

(e) Find the total time from when the balls are released until  $P$  first comes to rest.

(4)

TOPIC

# Moments

## Question 7

## Moments

7.

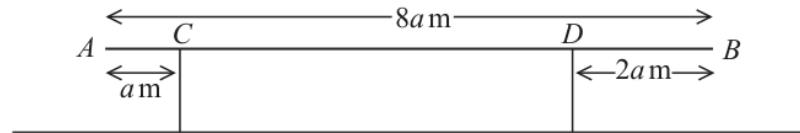


Figure 2

A non-uniform beam  $AB$ , of mass  $60$  kg and length  $8a$  metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at  $C$ , where  $AC = a$  metres and the other support is at  $D$ , where  $DB = 2a$  metres, as shown in Figure 2.

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By modelling the beam as a non-uniform rod whose centre of mass is at a distance  $x$  metres from  $A$ ,

- (a) find an expression for  $x$  in terms of  $a$ .

(5)

A box of mass  $M$  kg is placed on the beam at  $E$ , where  $AE = 2a$  metres.

The beam remains in equilibrium in a horizontal position.

The magnitude of the normal reaction between the beam and the support at  $C$  is now equal to the magnitude of the normal reaction between the beam and the support at  $D$ .

By modelling the box as a particle,

- (b) find the value of  $M$ .

(5)